

# 1 Rings and Ideals

## IDEALS. PRIME AND MAXIMAL IDEALS

**Proposition 1.1.** *There is a one-to-one order-preserving correspondence between the ideals  $\mathfrak{b}$  of  $A$  which contain  $\mathfrak{a}$ , and the ideals  $\bar{\mathfrak{b}}$  of  $A/\mathfrak{a}$ , given by  $\mathfrak{b} = \phi^{-1}(\bar{\mathfrak{b}})$ .*

**Corollary 1.5.** *Every non-unit of  $A$  is contained in a maximal ideal.*

**Corollary 1.6.** (i) *Let  $A$  be a ring and  $\mathfrak{m} \neq (1)$  an ideal of  $A$  such that every  $x \in A - \mathfrak{m}$  is a unit in  $A$ , Then  $A$  is a local ring and  $\mathfrak{m}$  its maximal ideal.*

(ii) *Let  $A$  be a ring and  $\mathfrak{m}$  a maximal ideal of  $A$ , such that every element of  $1 + \mathfrak{m}$  is a unit in  $A$ , then  $A$  is a local ring.*

## NILRADICAL AND JACOBSON RADICAL

We denote nilradical to be  $\mathfrak{N}$  and Jacobson radical to be  $\mathfrak{J}$ .

**Proposition 1.8.** *The nilradical of  $A$  is the intersection of all the prime ideals of  $A$ .*

*Sketch proof.* Assume that  $f \notin \mathfrak{N}$ . Let  $\Sigma$  be a set of ideals  $\mathfrak{a}$  with property  $f^n \notin \mathfrak{a}$ . Then  $\Sigma$  has maximal element which is prime.  $\square$

**Proposition 1.9.**  $x \in \mathfrak{J} \Leftrightarrow 1 - xy$  is a unit in  $A$  for all  $y \in A$ .

## OPERATIONS ON IDEALS

Two ideals  $\mathfrak{a}, \mathfrak{b}$  are said to be **coprime** if  $\mathfrak{a} + \mathfrak{b} = 1$ . Thus we have  $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{a}\mathfrak{b}$ .

**Proposition 1.10** (Chinese Remainder Theorem). *Define  $\phi$  to be*

$$\phi : A \rightarrow \prod_{i=1}^n (A/\mathfrak{a}_i), \quad \phi(x) = (x + \mathfrak{a}_1, \dots, x + \mathfrak{a}_n).$$

(i) *If  $\mathfrak{a}_i$  and  $\mathfrak{a}_j$  are coprime whenever  $i \neq j$ , then  $\mathfrak{a}_1 \cdots \mathfrak{a}_n = \bigcap_{i=1}^n \mathfrak{a}_i$ .*

(ii)  *$\phi$  is surjective  $\Leftrightarrow \mathfrak{a}_i$  and  $\mathfrak{a}_j$  are coprime whenever  $i \neq j$ .*

(iii)  *$\phi$  is injective  $\Leftrightarrow \bigcap_{i=1}^n \mathfrak{a}_i = (0)$ .*

**Proposition 1.11.** *Denote  $\mathfrak{a}$  to be ideal and  $\mathfrak{p}$  to be prime ideal.*

(i) *If  $\mathfrak{a} \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$ , then  $\mathfrak{a} \subseteq \mathfrak{p}_i$  for some  $i$ .*

(ii) *If  $\mathfrak{p} \supseteq \bigcap_{i=1}^n \mathfrak{a}_i$ , then  $\mathfrak{p} \supseteq \mathfrak{a}_i$  for some  $i$ . If  $\mathfrak{p} = \bigcap_{i=1}^n \mathfrak{a}_i$ , then  $\mathfrak{p} = \mathfrak{a}_i$  for some  $i$ .*

The **ideal quotient** of  $\mathfrak{a}$  and  $\mathfrak{b}$  is

$$(\mathfrak{a} : \mathfrak{b}) = \{x \in A : x\mathfrak{b} \subseteq \mathfrak{a}\}.$$

The annihilator of  $\mathfrak{b}$  is  $\text{Ann}(\mathfrak{b}) = (0 : \mathfrak{b})$ .

The **radical** of  $\mathfrak{a}$  is

$$\sqrt{\mathfrak{a}} = \{x \in A : x^n \in \mathfrak{a} \text{ for some } n > 0\}.$$

## EXTENSION AND CONTRACTION

Let  $f : A \rightarrow B$  be a ring homomorphism. Let  $\mathfrak{a} \in A$  and  $\mathfrak{b} \in B$ . Define

$$\mathfrak{a}^e = Bf(\mathfrak{a}), \quad \mathfrak{b}^c = f^{-1}(\mathfrak{b})$$

to be the **extension** of  $\mathfrak{a}$  and the **contraction** of  $\mathfrak{b}$ , respectively. If  $\mathfrak{b}$  is prime,  $\mathfrak{b}^c$  is prime; if  $\mathfrak{a}$  is prime,  $\mathfrak{a}^e$  need not be prime.

**Proposition 1.17.**  $\mathfrak{a} \subseteq \mathfrak{a}^{ec}$ ,  $\mathfrak{b} \supseteq \mathfrak{b}^{ce}$ .

## DEFINITIONS AND PROPOSITIONS FROM EXERCISES

**Idempotent** is an element  $e$  which satisfies  $e^2 = e$ .

A ring in which every element is an idempotent is a **Boolean ring**.

A topological space  $X$  is **reducible** if it can be written as a union  $X = X_1 \cup X_2$  of two closed proper subsets  $X_1, X_2$  of  $X$ . Equivalently,  $X$  is **irreducible** if every pair of non-empty open sets in  $X$  intersect, or if every non-empty open set is dense in  $X$ .

Let  $A$  be a ring and let  $\text{Spec}(A)$  be the set of all prime ideals of  $A$ . For each subset  $E$  of  $A$ , let  $V(E)$  denote the set of all prime ideals of  $A$  which contain  $E$ . By [Exercise 1.15](#), the sets  $V(E)$  satisfy the axioms for closed sets in a topological space. The resulting topology on  $\text{Spec}(A)$  is called **Zariski topology**, and  $\text{Spec}(A)$  is called the **prime spectrum** of  $A$ .

For each  $f \in A$ , the complement of  $V(f)$  in  $\text{Spec}(A)$ , denote to be  $X_f$ , is called **basic open set** of  $\text{Spec}(A)$ .

The subspace of  $\text{Spec}(A)$  consisting of the maximal ideals of  $A$  with the induced topology, is called the **maximal spectrum** of  $A$  and is denoted by  $\text{Max}(A)$ .

**Proposition ([Exercise 1.27](#)).** *Let  $k$  be algebraically closed. Then the affine space  $k^n$  with its classical Zariski topology is homeomorphic to  $\text{Max}(k[x_1, \dots, x_n])$ .*

## 2 Modules

**Proposition 2.1.** (i) If  $L \supseteq M \supseteq N$  are  $A$ -modules, then  $(L/N)/(M/N) \cong L/M$ .

(ii) If  $M_1, M_2$  are submodules of  $M$ , then  $(M_1 + M_2)/M_1 \cong M_2/(M_1 \cap M_2)$ .

An  $A$ -module is **faithful** if  $\text{Ann}(M) = 0$ .

### FINITELY GENERATED MODULES

**Proposition 2.3.**  $M$  is a finitely generated  $A$ -module  $\Leftrightarrow M$  is isomorphic to a quotient of  $A^n$  for some integer  $n > 0$ . More precisely,  $n$  is the number of generators of  $M$  and may not be unique.

**Proposition 2.4** (**Cayley–Hamilton theorem**). Let  $M$  be a finitely generated  $A$ -module, let  $\mathfrak{a}$  be an ideal of  $A$ , and let  $\phi$  be an  $A$ -module endomorphism of  $M$  such that  $\phi(M) \subseteq \mathfrak{a}M$ . Then  $\phi$  satisfies an equation of the form

$$\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$$

where the  $a_i$  are in  $\mathfrak{a}$ .

**Proposition 2.6** (**Nakayama's lemma**). Let  $M$  be a finitely generated  $A$ -module and  $\mathfrak{a}$  an ideal of  $A$  contained in the Jacobson radical of  $A$ . Then  $\mathfrak{a}M = M$  implies  $M = 0$ .

### EXACT SEQUENCES

**Proposition 2.10** (Snake lemma). Let

$$\begin{array}{ccccccc} 0 & \longrightarrow & M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' \longrightarrow 0 \\ & & \downarrow f' & & \downarrow f & & \downarrow f'' \\ 0 & \longrightarrow & N' & \xrightarrow{u'} & N & \xrightarrow{v'} & N'' \longrightarrow 0 \end{array}$$

be a commutative diagram of  $A$ -modules and homomorphisms, with the rows exact. Then there exists an exact sequence

$$0 \longrightarrow \ker(f') \xrightarrow{\bar{u}} \ker(f) \xrightarrow{\bar{v}} \ker(f'') \xrightarrow{\delta} \text{coker}(f') \xrightarrow{\bar{u}'} \text{coker}(f) \xrightarrow{\bar{v}'} \text{coker}(f'') \longrightarrow 0.$$

Let  $C$  be a class of  $A$ -modules and let  $\lambda$  be a function on  $C$  with values in  $\mathbb{Z}$ . The function  $\lambda$  is **additive** if for each short exact sequence

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

in which all terms belong to  $C$ , we have  $\lambda(M') - \lambda(M) + \lambda(M'') = 0$ .

### TENSOR PRODUCT

**Proposition 2.12** (Universal property). Let  $M, N$  be  $A$ -modules. The tensor product of  $M$  and  $N$  over  $A$  is an  $A$ -module  $M \otimes_A N$  together with an  $A$ -bilinear map

$$f : M \times N \rightarrow M \otimes_A N, \quad (m, n) \mapsto m \otimes n,$$

such that, for every  $A$ -module  $P$  and every  $A$ -bilinear map  $\tau : M \times N \rightarrow P$ , there exists a unique  $A$ -linear map  $\tilde{\tau} : M \otimes_A N \rightarrow P$  such that  $\tau = \tilde{\tau} \circ f$ . Equivalently, the following diagram commutes:

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & M \otimes_A N \\ & \searrow \tau & \downarrow \tilde{\tau} \\ & & P \end{array}$$

**Proposition 2.15.** Let  $A, B$  be rings, let  $M$  be an  $A$ -module,  $P$  a  $B$ -module and  $N$  an  $(A, B)$ -bimodule. then

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P).$$

Let  $M$  be an  $A$ -module and  $f : A \rightarrow B$  be a homomorphism of rings. Define  $M_B = B \otimes_A M$ .

**Proposition 2.19.** For an  $A$ -module  $N$ , the following are equivalent:

- (i) The functor  $T_N : M \mapsto M \otimes_A N$  on the category of  $A$ -modules is exact.
- (ii) If  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  is any exact sequence of  $A$ -modules, then  $0 \rightarrow M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$  is exact.
- (iii) If  $f : M' \rightarrow M$  is injective, then  $f \otimes 1 : M' \otimes N \rightarrow M \otimes N$  is injective.
- (iv) If  $f : M' \rightarrow M$  is injective and  $M, M'$  are finitely generated, then  $f \otimes 1 : M' \otimes N \rightarrow M \otimes N$  is injective.

An  $A$ -module  $N$  satisfying these properties is said to be a **flat**  $A$ -module.

## ALGEBRA

Let  $f : A \rightarrow B$  be a ring homomorphism. If  $a \in A$  and  $b \in B$ , define a product

$$ab = f(a)b.$$

Thus,  $B$  has an  $A$ -module structure. The ring  $B$  is said to be an  **$A$ -algebra**.

A ring homomorphism  $f : A \rightarrow B$  is **finite**, if  $B$  is finitely generated  $A$ -module. We can also say  $B$  is a **finite**  $A$ -algebra, or  $B$  is **finite** over  $A$ .

The ring homomorphism  $f$  is **of finite type**, if  $B$  is a finitely generated  $A$ -algebra.

## DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A partially ordered set  $I$  is said to be a **directed** set if for each pair  $i, j \in I$  there exists  $k \in I$  such that  $i \leq k$  and  $j \leq k$ .

Let  $A$  be a ring, let  $I$  be a directed set and let  $(M_i)_{i \in I}$  be a family of  $A$ -modules indexed by  $I$ . For each pair  $i, j \in I$  such that  $i \leq j$ , let  $\mu_{ij} : M_i \rightarrow M_j$  be an  $A$ -homomorphism, and suppose:

- (1)  $\mu_{ii}$  is the identity mapping of  $M_i$ , for all  $i \in I$ ;
- (2)  $\mu_{ik} = \mu_{jk} \circ \mu_{ij}$  whenever  $i \leq j \leq k$ .

Then the modules  $M_i$  and homomorphisms  $\mu_{ij}$  are said to form a **direct system**  $\mathbf{M} = (M_i, \mu_{ij})$  over the directed set  $I$ .

Let  $D$  be the submodule of  $\bigoplus M_i$  generated by all elements of the form  $x_i - \mu_{ij}(x_i)$  whenever  $i \leq j$  and  $x_i \in M_i$ . The **direct limit** of the direct system  $\mathbf{M}$  is defined to be

$$\varinjlim M_i = \bigoplus_{i \in I} M_i / D.$$

Let  $\mu : \bigoplus M_i \rightarrow \varinjlim M_i$  be the projection of quotient module. Define  $\mu_i$  to be the restriction of  $\mu$  to  $M_i$ .

A ring  $A$  (not necessarily commutative) is **von Neumann regular** if for every  $a \in A$ , there is an  $x \in A$  with  $a = axa$ . A commutative ring  $A$  is **absolutely flat** if every  $A$ -module is flat. By [Exercise 2.27](#), absolute flat and von Neumann regular are identical properties.

# 3 Rings and Modules of Fractions

## MULTIPLICATIVELY CLOSED SETS

**Proposition 3.3.** *The operation  $S^{-1}$  is exact.*

**Proposition 3.5.** *Let  $M$  be an  $A$ -module. Then there exists a unique  $S^{-1}A$  module isomorphism  $f : S^{-1}A \otimes_A M \rightarrow S^{-1}M$ .*

## LOCAL PROPERTIES

Let  $P$  be a property of a ring  $A$  (or of an  $A$ -module  $M$ ). Then the property  $P$  is said to be a **local property**, if  $A$  (or  $M$ ) has  $P \Leftrightarrow A_{\mathfrak{p}}$  (or  $M_{\mathfrak{p}}$ ) has  $P$  for each prime ideal  $\mathfrak{p}$  of  $A$ .

**Proposition 3.8.** *Let  $M, N$  be  $A$ -modules. Then the following are equivalent:*

- (i)  $M = 0$ ;
- (ii)  $M_{\mathfrak{p}} = 0$  for all prime ideals  $\mathfrak{p}$  of  $A$ ;
- (iii)  $M_{\mathfrak{m}} = 0$  for all maximal ideals  $\mathfrak{m}$  of  $A$ .

**Proposition 3.9.** *Let  $\phi : M \rightarrow N$  be an  $A$ -module homomorphism. Then the following are equivalent:*

- (i)  $\phi$  is injective;
- (ii)  $\phi_{\mathfrak{p}} : M_{\mathfrak{p}} \rightarrow N_{\mathfrak{p}}$  is injective for each prime ideal  $\mathfrak{p}$ ;
- (iii)  $\phi_{\mathfrak{m}} : M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$  is injective for each prime ideal  $\mathfrak{m}$ .

Similarly with “injective” replaced by “surjective” throughout.

**Proposition 3.10.** *Let  $M$  be  $A$ -modules. Then the following are equivalent:*

- (i)  $M$  is a flat  $A$ -module;
- (ii)  $M_{\mathfrak{p}}$  is a flat  $A_{\mathfrak{p}}$ -module for each prime ideal  $\mathfrak{p}$ ;
- (iii)  $M_{\mathfrak{m}}$  is a flat  $A_{\mathfrak{m}}$ -module for each maximal ideal  $\mathfrak{m}$ .

## EXTENSIONS AND CONTRACTIONS OF FRACTIONS

If  $\mathfrak{a}$  is an ideal in  $A$ , its extension  $\mathfrak{a}^e$  in  $S^{-1}A$  is  $S^{-1}\mathfrak{a}$ . Denote the contraction of  $S^{-1}\mathfrak{a}$  in  $A$  to be  $S(\mathfrak{a})$ .

**Proposition 3.11.** (i) *Every ideal in  $S^{-1}A$  is an extended ideal.*

(ii) *If  $\mathfrak{a}$  is an ideal in  $A$ , then  $S(\mathfrak{a}) = \mathfrak{a}^{ec} = \bigcup_{s \in S} (\mathfrak{a} : s)$ .*

(iii)  *$\mathfrak{a}$  is a contracted ideal  $\Leftrightarrow$  no element of  $S$  is a zero-divisor in  $A/\mathfrak{a}$ .*

(iv) *Them prime ideal of  $S^{-1}A$  are in one-to-one correspondence ( $\mathfrak{p} \leftrightarrow S^{-1}\mathfrak{p}$ ) with the prime ideal of  $A$  which don't meet  $S$ .*

(v) *The operation  $S^{-1}$  commutes with formation of finite sums, products, intersections and radicals.*

**Corollary 3.15.** *If  $N, P$  are submodules of an  $A$ -module  $M$  and if  $P$  is finitely generated, then  $S^{-1}(N : P) = (S^{-1}N : S^{-1}P)$ .*

## DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A multiplicatively closed subset  $S$  of a ring  $A$  is said to be **saturated** if

$$xy \in S \Leftrightarrow x \in S \text{ and } y \in S.$$

Let  $M$  be an  $A$ -module. An element  $x \in M$  is a **torsion** element of  $M$  if there exist a non-zero-divisor  $a \in A$  such that  $ax = 0$ . All torsion elements of  $M$  form a submodule of  $M$ , which is called the **torsion submodule** of  $M$  and denote by  $T(M)$ . If  $T(M) = 0$ , the module  $M$  is said to be **torsion free**.

An  $A$ -module  $M$  is called **faithfully flat** if the tensor product functor  $- \otimes_A M$  is exact and faithful. See [Exercise 3.16](#).

Let  $M$  be an  $A$ -module. The **support** of  $M$  is defined to be the set  $\text{Supp}(M)$  of prime ideals  $\mathfrak{p}$  such that  $M_{\mathfrak{p}} \neq 0$ .

$\text{Spec}(A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \otimes_A B)$  is called the **fiber** of  $f^* : \text{Spec}(B) \rightarrow \text{Spec}(A)$  over  $\mathfrak{p}$ .

# 4 Primary Decomposition

## PRIMARY IDEALS

An ideal  $\mathfrak{q} \in A$  is **primary** if  $\mathfrak{q} \neq A$ , and if  $x \notin \mathfrak{q} \Rightarrow y^n \in \mathfrak{q}$  for some  $n > 0$ . This also means  $x \notin \sqrt{\mathfrak{q}} \Rightarrow y \in \mathfrak{q}$ .

**Proposition 4.4.** *Let  $\mathfrak{q}$  be a  $\mathfrak{p}$ -primary ideal,  $x \in A$  an element. Then*

- (i) *if  $x \in \mathfrak{q}$  then  $(\mathfrak{q} : x) = (1)$ ;*
- (ii) *if  $x \notin \mathfrak{q}$  then  $(\mathfrak{q} : x)$  is  $\mathfrak{p}$ -primary;*
- (iii) *if  $x \notin \mathfrak{p}$  then  $(\mathfrak{q} : x) = \mathfrak{q}$ ;*

**Proposition 4.8.** *Let  $S$  be a multiplicatively closed subset of  $A$ , and let  $\mathfrak{q}$  be a  $\mathfrak{p}$ -primary ideal.*

- (i) *If  $S \cap \mathfrak{p} \neq \emptyset$ , then  $S^{-1}\mathfrak{q} = S^{-1}A$ .*
- (ii) *If  $S \cap \mathfrak{p} = \emptyset$ , then  $S^{-1}\mathfrak{q}$  is  $S^{-1}\mathfrak{p}$ -primary and its contraction in  $A$  is  $\mathfrak{q}$ .*

A **primary decomposition** of an ideal  $\mathfrak{a}$  in  $A$  is an expression  $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ .

If  $\sqrt{\mathfrak{q}_i}$  are all distinct and  $\mathfrak{q}_i \not\supseteq \bigcap_{j \neq i} \mathfrak{q}_j$ , the primary decomposition is said to be **irredundant**.

## UNIQUENESS

**Proposition 4.5** (1st uniqueness theorem). *Let  $\mathfrak{a}$  be a decomposable ideal and let  $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$  be a minimal primary decomposition of  $\mathfrak{a}$ . Let  $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$ . Then  $\mathfrak{p}_i$  are precisely the prime ideals which occur in the set of ideals  $\sqrt{(\mathfrak{a} : x)}$ , and hence are independent of the particular decomposition of  $\mathfrak{a}$ .*

The prime ideals  $\mathfrak{p}_i$  are said to belong to  $\mathfrak{a}$ , or to be **associated** with  $\mathfrak{a}$ . The minimal elements of the set  $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$  are called the minimal or **isolated** prime ideals belonging to  $\mathfrak{a}$ . The others are called **embedded** prime ideals.

**Proposition 4.9.** *Let  $S$  be a multiplicatively closed subset of  $A$  and let  $\mathfrak{a}$  be a decomposable ideal. Let  $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$  be a minimal primary decomposition of  $\mathfrak{a}$ . Let  $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$  and suppose the  $\mathfrak{q}_i$  numbered so that  $S$  meets  $\mathfrak{p}_{m+1}, \dots, \mathfrak{p}_n$  but not  $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ . Then we have minimal primary decompositions*

$$S^{-1}\mathfrak{a} = \bigcap_{i=1}^m S^{-1}\mathfrak{q}_i, \quad S(\mathfrak{a}) = \bigcap_{i=1}^m \mathfrak{q}_i.$$

A set  $\Sigma$  of prime ideals belonging to  $\mathfrak{a}$  is said to be **isolated** if it satisfies the following condition: if  $\mathfrak{p}'$  is a prime ideal belonging to  $\mathfrak{a}$  and  $\mathfrak{p}' \subseteq \mathfrak{p}$  for some  $\mathfrak{p} \in \Sigma$ , then  $\mathfrak{p}' \in \Sigma$ . Thus, every isolated set contains an isolated prime ideal.

**Proposition 4.10** (2nd uniqueness theorem). *Let  $\mathfrak{a}$  be a decomposable ideal, let  $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$  be a minimal primary decomposition of  $\mathfrak{a}$ , and let  $\{\mathfrak{p}_{i_1}, \dots, \mathfrak{p}_{i_m}\}$  be an isolated set of prime ideals of  $\mathfrak{a}$ . Then the ideal  $\mathfrak{q}_{i_1} \cap \dots \cap \mathfrak{q}_{i_m}$  is independent of the decomposition.*

**Corollary 4.11.** *The isolated primary components are uniquely determined by  $\mathfrak{a}$ .*

## DEFINITIONS AND PROPOSITIONS FROM EXERCISES

Let  $A$  be a ring and  $\mathfrak{p}$  a prime ideal of  $A$ . The  $n$ th **symbolic power** of  $\mathfrak{p}$  is defined to be the ideal  $\mathfrak{p}^{(n)} = S_{\mathfrak{p}}(\mathfrak{p}^n)$ .



# 5 Integral Dependence and Valuations

## INTEGRAL DEPENDENCE

**Proposition 5.1.** *The following are equivalent:*

- (i)  $x \in B$  is integral over  $A$ ;
- (ii)  $A[x]$  is a finitely generated  $A$ -module;
- (iii)  $A[x]$  is contained in a subring  $C$  of  $B$  such that  $C$  is a finitely generated  $A$ -module;
- (iv) There exist a faithful  $A[x]$ -module  $M$  which is finitely generated as an  $A$ -module.

The set  $\bar{A}$  of elements of  $B$  which are integral over  $A$  is called the **integral closure** of  $A$  in  $B$ . If  $\bar{A} = A$ , then  $A$  is said to be **integrally closed in  $B$** . If  $\bar{A} = B$ , the ring  $B$  is said to be **integral over  $A$** .

## THE GOING-UP THEOREM

**Proposition 5.7.** *Let  $A \subseteq B$  be integral domains,  $B$  integral over  $A$ . Then  $B$  is a field iff  $A$  is a field.*

**Corollary 5.9.** *Let  $A \subseteq B$  be rings,  $B$  integral over  $A$ . Let  $\mathfrak{q}_1, \mathfrak{q}_2$  be prime ideals of  $B$  such that  $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$  and  $\mathfrak{q}_1^e = \mathfrak{q}_2^e = \mathfrak{p}$ . Then  $\mathfrak{q}_1 = \mathfrak{q}_2$ .*

**Theorem 5.10 (Lying over property).** *Let  $A \subseteq B$  be rings,  $B$  integral over  $A$ , and let  $\mathfrak{p}$  be a prime ideal of  $A$ . Then there exists a prime ideal  $\mathfrak{q}$  of  $B$  such that  $\mathfrak{q} \cap A = \mathfrak{p}$ .*

**Theorem 5.11 (Going-up theorem).** *Let  $A \subseteq B$  be rings,  $B$  integral over  $A$ . Let  $\mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_n$  be a chain of prime ideals of  $A$  and  $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_m$  ( $m < n$ ) a chain of prime ideals of  $B$  such that  $\mathfrak{q}_i \cap A = \mathfrak{p}_i$  for  $1 \leq i \leq m$ . Then the chain  $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_m$  can be extended to a chain  $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_n$  such that  $\mathfrak{q}_i \cap A = \mathfrak{p}_i$  for  $1 \leq i \leq n$ .*

## THE GOING-DOWN THEOREM

**Proposition 5.12.** *Let  $A \subseteq B$  be rings,  $C$  the integral closure of  $A$  in  $B$ . Let  $S$  be a multiplicatively closed subset of  $A$ . Then  $S^{-1}C$  is the integral closure of  $S^{-1}A$  in  $S^{-1}B$ .*

An integral domain is said to be **integrally closed** (without qualification) if it is integrally closed in its field of fractions.

**Proposition 5.13.** *Let  $A$  be an integral domain. Then the following are equivalent:*

- (i)  $A$  is integrally closed;
- (ii)  $A_{\mathfrak{p}}$  is integrally closed, for each prime ideal  $\mathfrak{p}$ ;
- (iii)  $A_{\mathfrak{m}}$  is integrally closed, for each maximal ideal  $\mathfrak{m}$ .

**Theorem 5.16 (Going-down theorem).** *Let  $A \subseteq B$  be integral domains,  $A$  integrally closed,  $B$  integral over  $A$ . Let  $\mathfrak{p}_1 \supseteq \cdots \supseteq \mathfrak{p}_n$  be a chain of prime ideals of  $A$  and  $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_m$  ( $m < n$ ) a chain of prime ideals of  $B$  such that  $\mathfrak{q}_i \cap A = \mathfrak{p}_i$  for  $1 \leq i \leq m$ . Then the chain  $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_m$  can be extended to a chain  $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_n$  such that  $\mathfrak{q}_i \cap A = \mathfrak{p}_i$  for  $1 \leq i \leq n$ .*

## VALUATION RINGS

Let  $B$  be an integral domain.  $B$  is a **valuation ring** of  $\text{Frac}(B)$  if, for each  $x \neq 0$ , either  $x \in B$  or  $x^{-1} \in B$  (or both).

**Proposition 5.18.** *Let  $B$  be a valuation ring,  $K = \text{Frac}(B)$  its field of fractions. Then*

- (i)  $B$  is a local ring;
- (ii) If  $B'$  is a ring such that  $B \subseteq B' \subseteq K$ , then  $B'$  is a valuation ring of  $K$ ;
- (iii)  $B$  is integrally closed.

**Theorem 5.21.** *Let  $K$  be a field,  $\Omega$  an algebraically closed field. Let  $\Sigma$  be the set of all pairs  $(A, f)$ , where  $A$  is a subring of  $K$  and  $f : A \rightarrow \Omega$  a homomorphism. Define a partial order of  $\Sigma$  to be:*

$$(A, f) \leq (B, g) \Leftrightarrow A \subseteq B \text{ and } g|_A = f.$$

*Then the maximal element of  $\Sigma$  is a valuation ring of  $K$ .*

## DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A ring homomorphism  $f : A \rightarrow B$  is said to have the **going-up property** if for  $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$  and  $\mathfrak{p}_1 = f^{-1}(\mathfrak{q}_1)$ , there exist a  $\mathfrak{q}_2$  such that  $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$  and  $f^{-1}(\mathfrak{q}_2) = \mathfrak{p}_2$ . Reversing the inclusions of prime ideals we have the **going-down property**.

**Proposition (Noether normalization lemma).** *Let  $k$  be a field and let  $A \neq 0$  be a finitely generated  $k$ -algebra. Then there exist elements  $y_1, \dots, y_r \in A$  which are algebraically independent over  $k$  and such that  $A$  is integral, and therefore finite over  $k[y_1, \dots, y_r]$ .*

A ring  $A$  satisfied the following equivalent property is a **Jacobson ring**.

- (i) Every prime ideal in  $A$  is an intersection of maximal ideals.
- (ii) In every homomorphic image of  $A$  the nilradical is equal to the Jacobson radical.
- (iii) Every prime ideal in  $A$  which is not maximal is equal to the intersection of the prime ideals which contain it strictly.

A subset of a topological space  $X$  is **locally closed** if it is the intersection of an open set and a closed set.

A subset  $X_0$  satisfying the following equivalent conditions is said to be **very dense** in  $X$ :

- (1) Every non-empty locally closed subset of  $X$  meets  $X_0$ .
- (2) For every closed set  $E$  in  $X$  we have  $\overline{E \cap X_0} = E$ .
- (3) The mapping  $U \mapsto U \cap X_0$  of the collection of open sets of  $X$  on to the collection of open sets of  $X_0$  is bijective.

Let  $A$  be a valuation ring of a field  $K$ . The group  $U$  of units of  $A$  is a subgroup of the multiplicative group  $K^*$  of  $K$ . Let  $\Gamma = K^*/U$ . If  $\xi, \iota \in \Gamma$  are represented by  $x, y \in K$ , define  $\xi \geq \iota$  to mean  $xy^{-1} \in A$ . Then  $\Gamma$  is a totally ordered abelian group. It is called the **value group** of  $A$ .

Conversely, let  $\Gamma$  be a totally ordered abelian group. A **valuation** of  $K$  with values in  $\Gamma$  is a mapping  $v : K^* \rightarrow \Gamma$  such that

- (1)  $v(xy) = v(x) + v(y)$ ,
- (2)  $v(x + y) \geq \min(v(x), v(y))$ ,

for all  $x, y \in K^*$ . Then the set of elements  $x \in K^*$  such that  $v(x) \geq 0$  is a valuation ring of  $K$ . This ring is called the **valuation ring** of  $v$ , and the subgroup  $v(K^*)$  of  $\Gamma$  is the **value group** of  $v$ . Thus the concepts of valuation ring and valuation are equivalent.

## 6 Chain Conditions

Let  $\Sigma$  be a partially ordered set.  $\Sigma$  said to be satisfying the **ascending chain condition** (a.c.c) if, for every increasing sequence  $x_1 \leq x_2 \leq \dots$  in  $\Sigma$  there exists  $n$  such that  $x_n = x_{n+1} = \dots$ .

Similarly,  $\Sigma$  said to be satisfying the **descending chain condition** (d.c.c) if, for every decreasing sequence  $x_1 \geq x_2 \geq \dots$  in  $\Sigma$  there exists  $n$  such that  $x_n = x_{n+1} = \dots$ .

A module satisfying the ascending chain condition is said to be **Noetherian**. A module satisfying the descending chain condition is said to be **Artinian**.

**Proposition 6.2.**  *$M$  is a Noetherian  $A$ -module  $\Leftrightarrow$  every submodule of  $M$  is finitely generated.*

**Proposition 6.3.** *Let  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  be an exact sequence of  $A$ -modules. Then*

(i)  *$M$  is Noetherian  $\Leftrightarrow M'$  and  $M''$  are Noetherian.*

(ii)  *$M$  is Artinian  $\Leftrightarrow M'$  and  $M''$  are Artinian.*

**Proposition 6.5.** *Let  $A$  be a Noetherian (resp. Artinian) ring,  $M$  a finitely generated  $A$ -module. Then  $M$  is Noetherian (resp. Artinian).*

A **chain** of submodules of a module  $M$  is a sequence  $(M_i)$  of submodules of  $M$  such that

$$M = M_0 \supsetneq M_1 \supsetneq \dots \supsetneq M_n = 0.$$

The **length** of the chain is  $n$ . A **composition series** of  $M$  is a maximal chain, that is one which no extra submodules can be inserted, or equivalently  $M_{i-1}/M_i$  is simple.

**Proposition 6.7.** *Suppose that  $M$  has a composition series of length  $n$ . Then every composition series of  $M$  has length  $n$ , and every chain can be extended to a composition series.*

The length of a composition series of  $M$  is called the **length** of  $M$ , denoted as  $l(M)$ .

**Proposition 6.10.** *For a vector space,*

$$\text{finite dimension} \iff \text{finite length} \iff \text{a.c.c.} \iff \text{d.c.c.}$$

*If the conditions are satisfied, length equals dimension.*

## DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A topological space  $X$  is said to be **Noetherian** if the open subsets of  $X$  satisfy the ascending chain condition. Or equivalently, the closed subsets satisfy descending chain condition.

## 7 Noetherian Rings

**Proposition 7.1.** *Any homomorphic image of Noetherian ring is Noetherian.*

**Proposition 7.2.** *Let  $A$  be a subring of  $B$ ; suppose that  $A$  is Noetherian and that  $B$  is finitely generated as an  $A$ -module. Then  $B$  is Noetherian.*

**Proposition 7.3.** *If  $A$  is Noetherian and  $S$  is any multiplicatively closed subset of  $A$ , then  $S^{-1}A$  is Noetherian.*

**Theorem 7.5 (Hilbert's basis theorem).** *If  $A$  is Noetherian, then the polynomial ring  $A[x]$  is Noetherian.*

**Proposition 7.8.** *Let  $A \subseteq B \subseteq C$  be rings. Suppose that  $A$  is Noetherian, that  $C$  is finitely generated as an  $A$ -algebra and that  $C$  is either (i) finitely generated as a  $B$ -module or (ii) integral over  $B$ . Then  $B$  is finitely generated as an  $A$ -algebra.*

**Proposition 7.9 (Zariski's lemma).** *Let  $k$  be a field,  $E$  a finitely generated  $k$ -algebra. If  $E$  is a field then it is a finite algebraic extension of  $k$ .*

### PRIMARY DECOMPOSITION IN NOETHERIAN RINGS

**Proposition 7.11.** *In a Noetherian ring every ideal is a finite intersection of irreducible ideals.*

**Proposition 7.12.** *In a Noetherian ring every irreducible ideal is primary.*

**Proposition 7.13.** *In a Noetherian ring every ideal has a primary decomposition.*

**Proposition 7.14.** *In a Noetherian ring every ideal contains a power of its radical.*

**Corollary 7.15.** *In a Noetherian ring the nilradical is nilpotent.*

**Corollary 7.16.** *Let  $A$  be a Noetherian ring,  $\mathfrak{m}$  a maximal ideal of  $A$ ,  $\mathfrak{q}$  any ideal of  $A$ . Then the following are equivalent:*

- (i)  $\mathfrak{q}$  is  $\mathfrak{m}$ -primary;
- (ii)  $\sqrt{\mathfrak{q}} = \mathfrak{m}$ ;
- (iii)  $\mathfrak{m}^n \subseteq \mathfrak{q} \subseteq \mathfrak{m}$  for some  $n > 0$ .

**Proposition 7.17.** *Let  $\mathfrak{a} \neq (1)$  be an ideal in a Noetherian ring. Then the prime ideals which belong to  $\mathfrak{a}$  are precisely the prime ideals which occur in the set of ideals  $(\mathfrak{a} : x)$ .*

### DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A subset of a topological space  $X$  is **constructible set** if it is a finite union of locally closed sets.

## 8 Artinian Rings

**Proposition 8.1.** *In an Artinian ring every prime ideal is maximal.*

**Proposition 8.3.** *An Artinian ring has only a finite number of maximal ideals.*

**Proposition 8.4.** *In an Artinian ring the nilradical is nilpotent.*

The **dimension** of  $A$  is the supremum of the length of all chains of prime ideals in  $A$ .

**Proposition 8.5.** *A ring  $A$  is Artinian  $\Leftrightarrow A$  is Noetherian and  $\dim(A) = 0$ .*

**Proposition 8.6.** *Let  $A$  be a Noetherian local ring,  $\mathfrak{m}$  its maximal ideal. Then exactly one of the following statements is true:*

- (i)  $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$  for all  $n$ ;
- (ii)  $\mathfrak{m}^n = 0$  for some  $n$ . In this case  $A$  is an Artinian local ring.

**Proposition 8.7** (Structure theorem of Artinian rings). *An Artinaian ring  $A$  is uniquely (up to isomorphism) a finite direct product of Artinian local rings.*

**Proposition 8.8.** *Let  $A$  be an Artinian local ring. Then the following are equivalent:*

- (i) *every ideal in  $A$  is principal;*
- (ii) *the maximal ideal  $\mathfrak{m}$  is principal;*
- (iii)  $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) \leq 1$ .

## 9 Discrete Valuation Rings and Dedekind Domain

**Proposition 9.1.** *Let  $A$  be a Noetherian domain of dimension 1. Then every non-zero ideal  $\mathfrak{a}$  in  $A$  can be uniquely expressed as a product of primary ideals whose radicals are all distinct.*

### DISCRETE VALUATION RINGS

Let  $K$  be a field. A **discrete valuation** on  $K$  is a mapping  $v$  of  $K^*$  onto  $\mathbb{Z}$  such that

- (1)  $v(xy) = v(x) + v(y)$ ;
- (2)  $v(x + y) \geq \min(v(x), v(y))$ .

The set consisting of all  $x \in K^*$  such that  $v(x) \geq 0$  called the **valuation ring** of  $v$ .

An integral domain  $A$  is a **discrete valuation ring** if there is a discrete valuation  $v$  of its field of fractions  $K$  such that  $A$  is the valuation ring of  $v$ .

**Proposition 9.2.** *Let  $A$  be a Noetherian local domain of dimension 1,  $\mathfrak{m}$  its maximal ideal. Then the following are equivalent:*

- (i)  $A$  is a discrete valuation ring;
- (ii)  $A$  is integrally closed;
- (iii)  $\mathfrak{m}$  is a principal ideal;
- (iv)  $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 1$ ;
- (v) Every non-zero ideal is a power of  $\mathfrak{m}$ ;
- (vi) There exists  $x \in A$  such that every non-zero ideal is of the form  $(x^k)$  for  $k \geq 0$ .

### DEDEKIND DOMAINS

**Theorem 9.3.** *Let  $A$  be a Noetherian local domain of dimension 1. Then the following are equivalent:*

- (i)  $A$  is integrally closed;
- (ii) Every primary ideal in  $A$  is a prime power;
- (iii) Every local ring  $A_{\mathfrak{p}}$  with  $\mathfrak{p} \neq 0$  is a discrete valuation ring.

A ring satisfying the conditions is called a **Dedekind domain**.

### FRACTIONAL IDEALS

Let  $A$  be an integral domain,  $K$  its field of fractions. An  $A$ -submodule  $M$  of  $K$  is a **fractional ideal** of  $A$  if  $xM \subseteq A$  for some  $x \neq 0$ .

An  $A$ -submodule  $M$  of  $K$  is an **invertible ideal** if there exists a submodule  $N$  of  $K$  such that  $MN = A$ . Then  $N = (A : M)$ , and  $M$  is finitely generated fractional ideal.

**Proposition 9.6.** *For a fractional ideal  $M$ , the following are equivalent:*

- (i)  $M$  is invertible;

(ii)  $M$  is finitely generated and , for each prime ideal  $\mathfrak{p}$ ,  $M_{\mathfrak{p}}$  is invertible;

(iii)  $M$  is finitely generated and , for each maximal ideal  $\mathfrak{m}$ ,  $M_{\mathfrak{m}}$  is invertible;

**Theorem 9.8.** *Let  $A$  be an integral domain. Then  $A$  is a Dedekind domain  $\Leftrightarrow$  every non-zero fractional ideal of  $A$  is invertible.*

# 10 Completion

## TOPOLOGIES AND COMPLETIONS

A **topological abelian group**  $G$  is both a topological space and an abelian group, with two structural mappings

$$\begin{aligned} G \times G &\rightarrow G : (x, y) \mapsto x + y \\ G &\rightarrow G : x \mapsto -x \end{aligned}$$

being continuous.

If  $\{0\}$  is closed in  $G$ , then  $G$  is Hausdorff.

Since  $T_a(x) = x + a$  is a homeomorphism, the topology of  $G$  is uniquely determined by the neighborhoods of 0 in  $G$ .

**Proposition 10.1.** *Let  $H$  be the intersection of all neighborhoods of 0 in  $G$ . Then*

- (i)  $H$  is a subgroup.
- (ii)  $H$  is the closure of  $\{0\}$ .
- (iii)  $G/H$  is Hausdorff.
- (iv)  $G$  is Hausdorff  $\Leftrightarrow H = 0$ .

Assume that  $0 \in G$  has a countable fundamental system of neighborhoods. A **Cauchy sequence**  $(x_\nu)$  of  $G$  is that, for any neighborhood  $U$  of 0, there exists an integer  $s(U)$  such that  $x_m - x_n \in U$  for all  $m, n \geq s(U)$ .

Let  $C(G)$  be the set of all Cauchy sequences in  $G$ . Define an equivalence relation  $\sim$  on  $C(G)$  where  $(x_\nu) \sim (y_\nu)$  means  $x_n - y_n \rightarrow 0$  in  $G$ . The **completion**  $\hat{G}$  of  $G$  is the set of all equivalence classes in  $C(G)/\sim$ .

For each  $x \in G$ , define a homomorphism  $\phi : G \rightarrow \hat{G}$  maps to a constant sequence  $(x)$ . Then  $\phi$  is injective iff  $G$  is Hausdorff.

Assume further that  $0 \in G$  has a fundamental system of neighborhoods consisting of a sequence of subgroups

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n \supseteq \cdots$$

For any  $(x_\nu) \in C(G)$  and a fixed  $n$ , the image sequence  $(\xi_\nu + G_n)$  in  $G/G_n$  is ultimately constant. Let  $\xi_n$  be this ultimate constant value.

As  $n$  varies, the projection  $\theta_{n+1} : G/G_{n+1} \rightarrow G/G_n$  satisfies  $\theta_{n+1}(\xi_{n+1}) = \xi_n$  for all  $n$ . The groups  $G/G_n$  combined with maps  $\theta_n$  form an **inverse system**, and the group of all such coherent sequences  $(\xi_\nu)$  is called the **inverse limit**, denoted as

$$\hat{G} = \varprojlim G/G_n.$$

A direct (or inverse) system is called **injective/surjective system**, if every transition map  $\mu_{ij}$  is injective/surjective.

**Proposition 10.2.** *If  $0 \rightarrow \{A_n\} \rightarrow \{B_n\} \rightarrow \{C_n\} \rightarrow 0$  is an exact sequence of inverse system, then*

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n$$

*is exact. If  $\{A_n\}$  is surjective, then*

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n \rightarrow 0$$

*is exact.*



Let  $A = \prod_{i=1}^{\infty} A_n$  and define  $d^A : A \rightarrow A$  by  $d^A(a_n) = a_n - \theta_{n+1}(a_{n+1})$ . Then  $\ker(d^A) \cong \varprojlim A_n$ . Similar to  $B$  and  $C$ , we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow d^A & & \downarrow d^B & & \downarrow d^C \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \end{array}$$

By [Snake lemma](#), we have an exact sequence

$$0 \longrightarrow \ker(d^A) \longrightarrow \ker(d^B) \longrightarrow \ker(d^C) \longrightarrow \operatorname{coker}(d^A) \longrightarrow \operatorname{coker}(d^B) \longrightarrow \operatorname{coker}(d^C) \longrightarrow 0.$$

and the  $\operatorname{coker}(d^A)$  is usually denoted by  $\varprojlim^1 A_n$ .

**Corollary 10.3.** *Let  $0 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 0$  be an exact sequence of groups. Let  $G$  have the topology defined by a sequence  $\{G_n\}$  of subgroups, and give  $G'$  and  $G''$  the topologies by the contraction and extension of  $\{G_n\}$ , respectively. Then*

$$0 \rightarrow \hat{G}' \rightarrow \hat{G} \rightarrow \hat{G}'' \rightarrow 0$$

*is exact.*

**Corollary 10.4.**  $\hat{G}_n$  is a subgroup of  $\hat{G}$  and  $\hat{G}/\hat{G}_n \cong G/G_n$ .

**Corollary 10.5.**  $\hat{\hat{G}}_n \cong \hat{G}_n$ .

If  $G \rightarrow \hat{G}$  is an isomorphism,  $G$  is said to be [complete](#).

Let  $A$  be a ring,  $\mathfrak{a}$  its ideal. The topology defined by  $G_n = \mathfrak{a}^n$  ( $G_0 = A$ ) is called the  [\$\mathfrak{a}\$ -adic topology](#), and the ring  $A$  combined with this topology is a [topological ring](#).

## FILTRATIONS

A chain  $M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_n \cdots$  is called a [filtration](#) of  $M$ , and denoted by  $(M_n)$ . It is an  [\$\mathfrak{a}\$ -filtration](#) if  $\mathfrak{a}M_n \subseteq M_{n+1}$  for all  $n$ , and a [stable  \$\mathfrak{a}\$ -filtration](#) if  $\mathfrak{a}M_n = M_{n+1}$  for all sufficiently large  $n$ .

The  [\$\mathfrak{a}\$ -adic completion](#) of  $M$  is defined to be

$$\hat{M} = \varprojlim M/\mathfrak{a}^n M.$$

## GRADED RINGS AND MODULES

A [graded ring](#) is a ring  $A$  together with a family  $(A_n)_{n \geq 0}$  of additive subgroups of  $A$ , such that  $A = \bigoplus_{n=0}^{\infty} A_n$  and  $A_m A_n \subseteq A_{m+n}$  for all  $m, n \geq 0$ .

**Proposition 10.7.** *A graded ring  $A$  is Noetherian, iff  $A_0$  is Noetherian and  $A$  is finitely generated as an  $A_0$ -algebra.*

**Proposition 10.9** (Artin-Rees lemma). *Let  $A$  be a Noetherian ring,  $\mathfrak{a}$  an ideal in  $A$ ,  $M$  a finitely-generated  $A$ -module,  $(M_n)$  a stable  $\mathfrak{a}$ -filtration of  $M$ . If  $M'$  is a submodule of  $M$ , then  $(M' \cap M_n)$  is a stable  $\mathfrak{a}$ -filtration of  $M'$ .*

**Proposition 10.12.** *Let  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  be an exact sequence of finitely-generated modules over a Noetherian ring  $A$ , then the sequence of  $\mathfrak{a}$ -adic completion  $0 \rightarrow \hat{M}' \rightarrow \hat{M} \rightarrow \hat{M}'' \rightarrow 0$  is exact.*

**Proposition 10.13.** *Let  $f : \hat{A} \otimes_A M \rightarrow \hat{A} \otimes_A \hat{M} \rightarrow \hat{A} \otimes_{\hat{A}} \hat{M} \rightarrow \hat{M}$ . If  $M$  is finitely-generated  $A$ -module,  $f$  is surjective. If moreover  $A$  is Noetherian,  $f$  is an isomorphism.*

**Proposition 10.14.** *If  $A$  is a Noetherian ring, the  $\mathfrak{a}$ -adic completion  $\hat{A}$  is a flat  $A$ -algebra.*

**Proposition 10.15.** *If  $A$  is a Noetherian ring,  $\hat{A}$  its  $\mathfrak{a}$ -adic completion, then*

- (i)  $\hat{\mathfrak{a}} = \hat{A}\mathfrak{a} \cong \hat{A} \otimes_A \mathfrak{a}$ ;
- (ii)  $\widehat{(\mathfrak{a}^n)} = (\hat{\mathfrak{a}})^n$ ;
- (iii)  $\hat{\mathfrak{a}}^n / \hat{\mathfrak{a}}^{n+1} \cong \mathfrak{a}^n / \mathfrak{a}^{n+1}$ ;
- (iv)  $\hat{\mathfrak{a}}$  is contained in the Jacobson radical of  $\hat{A}$ .

**Proposition 10.16.** *Let  $A$  be a Noetherian local ring,  $\mathfrak{m}$  its maximal ideal. Then the  $\mathfrak{m}$ -adic completion  $\hat{A}$  of  $A$  is a local ring with maximal ideal  $\hat{\mathfrak{m}}$ .*

**Proposition 10.17 (Krull intersection theorem).** *Let  $A$  be a Noetherian ring,  $\mathfrak{a}$  an ideal,  $M$  a finitely-generated  $A$ -module and  $\hat{M}$  the  $\mathfrak{a}$ -adic completion of  $M$ . Then the kernel  $\bigcap_{n=1}^{\infty} \mathfrak{a}^n M$  of  $M \rightarrow \hat{M}$  consists of all  $x \in M$  annihilated by some element of  $1 + \mathfrak{a}$ .*

## THE ASSOCIATED GRADED RING

Let  $A$  be a ring and  $\mathfrak{a}$  an ideal of  $A$ . The **associated graded ring** is the graded ring

$$G(A) (= G_{\mathfrak{a}}(A)) = \bigoplus_{n=0}^{\infty} \mathfrak{a}^n / \mathfrak{a}^{n+1} \quad (a^0 = A).$$

Similarly, if  $M$  is an  $A$ -module and  $(M_n)$  is an  $\mathfrak{a}$ -filtration of  $M$ , then the **associated graded module** is

$$G(M) = \bigoplus_{n=0}^{\infty} M_n / M_{n+1},$$

which is a graded  $G(A)$ -module. Let  $G_n(M) = M_n / M_{n+1}$ .

**Proposition 10.22.** *Let  $A$  be a Noetherian ring,  $\mathfrak{a}$  an ideal of  $A$ . Then*

- (i)  $G_{\mathfrak{a}}(A)$  is Noetherian;
- (ii)  $G_{\mathfrak{a}}(A)$  and  $G_{\hat{\mathfrak{a}}}(\hat{A})$  are isomorphic as graded rings;
- (iii) If  $M$  is a finitely-generated  $A$ -module and  $(M_n)$  is a stable  $\mathfrak{a}$ -filtration of  $M$ , then  $G(M)$  is a finitely-generated graded  $G_{\mathfrak{a}}(A)$ -module.

**Proposition 10.23.** *Let  $\phi : A \rightarrow B$  be a homomorphism of filtered groups, i.e.  $\phi(A_n) \subseteq B_n$ , and let  $G(\phi) : G(A) \rightarrow G(B)$ ,  $\hat{\phi} : \hat{A} \rightarrow \hat{B}$  be the induced homomorphisms of the associated graded and completed groups. Then*

- (i)  $G(\phi)$  injective  $\Rightarrow \hat{\phi}$  injective;
- (ii)  $G(\phi)$  surjective  $\Rightarrow \hat{\phi}$  surjective.

**Proposition 10.24.** *Let  $A$  be a ring,  $\mathfrak{a}$  an ideal of  $A$ ,  $M$  an  $A$ -module,  $(M_n)$  an  $\mathfrak{a}$ -filtration of  $M$ . Suppose that  $A$  is complete in the  $\mathfrak{a}$ -topology and that  $\bigcap_n M_n = 0$ . Suppose also  $G(M)$  is a finitely generated  $G(A)$ -module. Then  $M$  is a finitely generated  $A$ -module.*

**Theorem 10.26.** *If  $A$  is a Noetherian ring,  $\mathfrak{a}$  an ideal of  $A$ , then the  $\mathfrak{a}$ -adic completion  $\hat{A}$  of  $A$  is Noetherian.*

## DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A Noetherian topological ring in which the topology is defined by an ideal contained in the Jacobson radical is called a **Zariski ring**.

**Proposition (Hensel's lemma).** *Let  $A$  be a local ring,  $\mathfrak{m}$  its maximal ideal. Assume that  $A$  is  $\mathfrak{m}$ -adically complete. If  $f \in A[x]$  is monic of degree  $n$ , and if there exist coprime monic polynomials  $\bar{g}, \bar{h} \in (A/\mathfrak{m})[x]$  of degrees  $r, n-r$ , with  $\bar{f} = \bar{g}\bar{h}$ , then we can lift  $\bar{g}, \bar{h}$  back to monic polynomials  $g, h \in A[x]$  such that  $f = gh$ .*

# 11 Dimension Theory

## HILBERT FUNCTIONS

Let  $A$  be a Noetherian graded ring and  $M$  be a finitely-generated graded  $A$ -module. Let  $\lambda$  be an additive function on the class of all finitely generated  $A_0$ -modules. The **Poincaré series** of  $M$  is the power series

$$P(M, t) = \sum_{n=0}^{\infty} \lambda(M_n) t^n \in \mathbb{Z}[[t]].$$

**Theorem 11.1 (Hilbert-Serre theorem).**  $P(M, t)$  is a rational function in  $t$  of the form  $f(t) / \prod_{i=1}^d (1 - t^{k_i})$ , where  $f(t) \in \mathbb{Z}[[t]]$ .

The order of the pole of  $P(M, t)$  at  $t = 1$  is denoted as  $d(M)$ .

**Corollary 11.2.** If each  $k_i = 1$ , then for all sufficient large  $n$ ,  $\lambda(M_n)$  is a polynomial in  $n$ . More specifically, if  $f(t) = \sum_{k=0}^N a_k t^k$ , we have

$$\lambda(M_n) = \sum_{k=0}^n a_k \binom{d+n-k-1}{d-1}.$$

This is called **Hilbert function** or **Hilbert polynomial** of  $M$ , and its leading term is  $(\sum a_k) n^{d-1} / (d-1)!$ .

**Proposition 11.3.** If  $x \in A_k$  is not a zero-divisor in  $M$  then  $d(M/xM) = d(M) - 1$ .

**Proposition 11.4.** Let  $A$  be a Noetherian local ring,  $\mathfrak{m}$  its maximal ideal,  $\mathfrak{q}$  an  $\mathfrak{m}$ -primary ideal,  $M$  a finitely-generated  $A$ -module,  $(M_n)$  a stable  $\mathfrak{q}$ -filtration of  $M$ . Then

- (i)  $M/M_n$  is of finite length for each  $n \geq 0$ ;
- (ii) for all sufficiently large  $n$  this length is a polynomial  $\chi_{\mathfrak{q}}^M(n)$  of degree  $\leq s$ , where  $s$  is the least number of generators of  $\mathfrak{q}$ ;
- (iii) the degree and leading coefficient of  $\chi_{\mathfrak{q}}^M(n)$  depend only on  $M$  and  $\mathfrak{q}$ , not on the filtration chosen.

If  $M = A$ , the polynomial  $\chi_{\mathfrak{q}}^M(n)$  is called the **characteristic polynomial** of the  $\mathfrak{m}$ -primary ideal  $\mathfrak{q}$ , simply noted as  $\chi_{\mathfrak{q}}(n)$ .

**Proposition 11.6.**  $\deg \chi_{\mathfrak{q}}(n) = \deg \chi_{\mathfrak{m}}(n)$ .

If  $A$  is a ring,  $\mathfrak{p}$  a prime ideal in  $A$ , then the **height** of  $\mathfrak{p}$  is defined to be the supremum of chains of prime ideals  $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r = \mathfrak{p}$  which end at  $\mathfrak{p}$ . Hence  $\text{height } \mathfrak{p} = \dim A_{\mathfrak{p}}$ .

Likewise we define the **depth** of  $\mathfrak{p}$ , by considering chains of prime ideals which start at  $\mathfrak{p}$ . Hence,  $\text{depth } \mathfrak{p} = \dim A/\mathfrak{p}$ .

## DIMENSION THEORY OF NOETHERIAN LOCAL RINGS

**Theorem 11.14 (Dimension theorem).** For any Noetherian local ring  $A$  the following three integers are equal:

- (i) the maximum length of chains of prime ideals in  $A$  (**Krull dimension**);
- (ii) the degree of the characteristic polynomial  $\chi_{\mathfrak{m}} = l(A/\mathfrak{m}^n)$ ;

(iii) the least number of generators of an  $\mathfrak{m}$ -primary ideal of  $A$ .

**Corollary 11.17** (Krull's principal ideal theorem). *Let  $A$  be a Noetherian ring and let  $x$  be an element of  $A$  which is neither a zero-divisor nor a unit. Then every minimal prime ideal  $\mathfrak{p}$  of  $(x)$  has height 1.*

**Corollary 11.18.** *Let  $A$  be a Noetherian local ring,  $x$  an element of  $\mathfrak{m}$  which is not a zero-divisor. Then  $\dim A/(x) = \dim A - 1$ .*

**Corollary 11.19.** *Let  $\hat{A}$  be the  $\mathfrak{m}$ -adic completion of  $A$ . Then  $\dim A = \dim \hat{A}$ .*

If  $x_1, \dots, x_d$  generate an  $\mathfrak{m}$ -primary ideal, and  $d = \dim A$ , we call  $x_1, \dots, x_d$  a **system of parameters**.

**Proposition 11.20.** *Let  $x_1, \dots, x_d$  be a system of parameters for  $A$  and let  $\mathfrak{q} = (x_1, \dots, x_d)$  be the  $\mathfrak{m}$ -primary ideal. Let  $f(t_1, \dots, t_d)$  be a homogeneous polynomial of degree  $s$  with coefficients in  $A$ , and assume that  $f(x_1, \dots, x_d) \in \mathfrak{q}^{s+1}$ . Then all the coefficients of  $f$  lie in  $\mathfrak{m}$ .*

## REGULAR LOCAL RINGS

**Theorem 11.22.** *Let  $A$  be a Noetherian local ring of dimension  $d$ ,  $\mathfrak{m}$  its maximal ideal,  $k = A/\mathfrak{m}$ . Then the following are equivalent:*

- (i)  $G_{\mathfrak{m}}(A) \cong k[t_1, \dots, t_d]$  where the  $t_i$  are independent indeterminates;
- (ii)  $\dim_k (\mathfrak{m}/\mathfrak{m}^2) = d$ ;
- (iii)  $\mathfrak{m}$  can be generated by  $d$  elements.

The ring satisfying any of the conditions is called **regular** local ring.

**Proposition 11.23.** *Let  $A$  be a ring,  $\mathfrak{a}$  an ideal of  $A$  such that  $\bigcap_n \mathfrak{a}^n = 0$ . Suppose that  $G_{\mathfrak{a}}(A)$  is an integral domain. Then  $A$  is an integral domain.*

**Proposition 11.24.** *Let  $A$  be a Noetherian local ring. Then  $A$  is regular iff  $\hat{A}$  is regular.*

## TRANSCENDENTAL DIMENSION

Assume that  $k$  is an algebraically closed field and let  $V$  be an irreducible affine variety over  $k$ . Let the coordinate ring of  $V$  be  $A(V) = k[x_1, \dots, x_n]/I(V)$ . Then  $k(V) = \text{Frac } A(V)$  is a finitely generated extension of  $k$  and so has a finite transcendence degree over  $k$ , which is defined to be the **dimension** of  $V$ .

If  $P$  is a point with maximal ideal  $\mathfrak{m}$  we call the Krull dimension  $\dim A(V)_{\mathfrak{m}}$  the **local dimension** of  $V$  at  $P$ .

**Theorem 11.25.** *For any irreducible variety  $V$  over  $k$  the local dimension at any point is equal to its dimension.*

# EXERCISES

## 1 Rings and Ideals

1.1. Consider  $(1+x)(1-x+x^2-\cdots+(-x)^{n-1})$ .

1.2. (i) Passing to  $(A/\mathfrak{p})[x]$ .  $A/\mathfrak{p}$  is an integral domain, so  $\deg(\overline{f\bar{g}}) = \deg(\overline{f}) + \deg(\overline{g})$ .  $\overline{f\bar{g}} = 1$  means  $a_1, \dots, a_n \in \mathfrak{p}$ . Use [Proposition 1.8](#).

(ii)  $a, b$  are nilpotents  $\Rightarrow a+b$  is nilpotent. Note that  $f-a_0 = xf_1$  is nilpotent.

(iii) Let  $gf=0$  and  $g$  is of minimal degree. The leading coefficient  $a_nb_m=0$ , so  $a_ng$  has less degree than  $g$  and so  $a_ng=0$ . Same process to the second leading and following coefficients, we have  $a_i g=0$  for all  $i$ .

(iv) (Gauss's lemma) Let  $\mathfrak{a}, \mathfrak{b}$  and  $\mathfrak{c}$  are generated by the coefficients of  $f, g$  and  $fg$ , respectively. We need to proof  $\mathfrak{c} = (1) \Leftrightarrow \mathfrak{a}\mathfrak{b} = (1)$ .  $\mathfrak{c} \subseteq \mathfrak{a}\mathfrak{b}$  by the form of coefficients in  $fg$ . If  $\mathfrak{c} \subsetneq \mathfrak{a}\mathfrak{b} = (1)$ , let  $\mathfrak{m}$  be the maximal ideal containing  $\mathfrak{c}$ , then  $(A/\mathfrak{m})[x]$  is an integer domain.  $\overline{f\bar{g}} = 0$  but  $\overline{f}, \overline{g} \neq 0$ .

1.3. Induction on the indeterminates.

1.4. Pick the  $y$  in [Proposition 1.9](#) to be the indeterminate, then use [Exercise 1.2\(i\)](#).

1.5. (i)  $\Leftarrow$ :  $fg = \sum_{i=0}^{\infty} \left( \sum_{j=0}^i a_j b_{i-j} \right) x^i$ , so we can find  $g$  such that  $\sum_{j=0}^i a_j b_{i-j} = 0$ .

(ii) Similar to [Exercise 1.2\(ii\)](#). Construct a counterexample where  $a_n$  have increasing nilpotencies.

(iii) Pick the  $y$  in [Proposition 1.9](#) to be 1. Use (i) on  $1-f \Leftrightarrow 1-a_0$ .

(iv),(v)  $A \cong A[[x]]/(x)$ . Use [Proposition 1.1](#).

1.6.  $\mathfrak{J}$  is an ideal, so there is an idempotent  $e \in \mathfrak{J}$  but  $e \notin \mathfrak{R}$ . Pick the  $y$  in [Proposition 1.9](#) to be 1.

1.7. Note that  $x(1-x^{n-1}) = 0 \in \mathfrak{p}$ .

1.8. Zorn's lemma but define the partial order reversely.

1.9.  $\Rightarrow$ : Apply [Proposition 1.8](#) to  $A/\mathfrak{a}$ .

$\Leftarrow$ : Use [Proposition 1.11\(ii\)](#).

1.10. Use [Corollary 1.5](#) and [Corollary 1.6\(i\)](#)

1.11. (i)  $x+x = (x+x)^2 = x^2 + 2x + x^2 = x + 2x + x$ .

(ii) Proof that every element in  $A/\mathfrak{p}$  is 1.

(iii)  $x+y-xy$ .

1.12. Use [Corollary 1.5](#).

1.13. **1 is a monic polynomial but it is neither irreducible nor reducible.**  $A$  is an integral domain, so  $\deg(fg) = \deg(f) + \deg(g)$ . If  $\mathfrak{a} = (1)$ , apply the equation to  $1 = \sum_f y_f f(x_f)$ .

1.14. If  $xy \in \mathfrak{m}$ ,  $xy$  is zero-divisor, so are  $x$  and  $y$ . By maximality,  $\mathfrak{m} \subseteq \mathfrak{m} + (x) = \mathfrak{m}$ .

1.15. (i) Apply [Proposition 1.8](#) to  $A/\mathfrak{a}$  to proof the second equation.

(iv) Use  $\sqrt{\mathfrak{a} \cap \mathfrak{b}} = \sqrt{\mathfrak{a}\mathfrak{b}}$ .

1.16. Irreducible polynomials relate to prime ideals.

1.17. (i)-(iv) Use [Exercise 1.15](#).

(v) Let  $(X_{f_i})_{i \in I}$  be an open cover. By [Exercise 1.15](#),  $V(\{f_i\}_{i \in I}) = \emptyset$ , so  $\{f_i\}_{i \in I}$  generates the unit ideal. Let  $1 = \sum_{i \in J} g_i f_i$  where  $J$  is a finite subset of  $I$ . If  $\mathfrak{p} \notin \bigcup_{i \in J} X_{f_i}$ , then  $f_i \in \mathfrak{p}$  for all  $i \in J$ .

## EXERCISES

(vi) Similar to (v), where the unit ideal shall change to the radical  $\sqrt{(f)}$ .

**1.18.** (iv) If  $V(\mathfrak{p}_x) \subseteq V(\mathfrak{p}_y)$ ,  $\mathfrak{p}_x \in \overline{V(\mathfrak{p}_y)}$ . If they don't have containing relation,  $\mathfrak{p}_x \in \overline{V(\mathfrak{p}_y)}$  and  $\mathfrak{p}_y \in \overline{V(\mathfrak{p}_x)}$ .

**1.19.** Use the closed sets definition. Consider minimal prime ideals.

**1.20.** (i) Let  $\overline{Y} = Z_1 \cup Z_2$ , then  $Z_i \cap Y = \emptyset$  for an  $i \in \{1, 2\}$ . Thus  $Z_i = \overline{Z_i \cap Y} = \emptyset$ .

(ii) Similar to (i). Construct the maximal element and proof that it is irreducible.

(iii) If  $Y$  Hausdorff but irreducible, every pair of open sets in  $Y$  intersect. Thus  $Y$  is singleton.

(iv) Use the closed set definition.

**1.21.** Simplify the sets:  $\phi^{*-1}(E) = \{\mathfrak{q} \in Y : \phi^{-1}(\mathfrak{q}) \in E\}$ ,  $\phi^*(V(E)) = \{\phi^{-1}(\mathfrak{q}) : E \subseteq \mathfrak{q}\}$ .

**1.22.** Use quotient to find the form of prime ideals in  $A$ .

**1.23.** (i)  $X_f = V(1 - f)$ .

(ii)  $X_{f_1} \cup X_{f_2} = X_{f_1 + f_2 - f_1 f_2}$ .

(iii) A closed set in compact space is compact. By [Exercise 1.17](#), it is a finite union of basic open set. Then apply (ii).

(iv) By [Exercise 1.11](#) every prime ideal is maximal. Pick  $f \in \mathfrak{p}_x \setminus \mathfrak{p}_y$ , then  $X_f$  and  $V(f)$  separate two points.

**1.24.**  $a = a \wedge 1 = a \wedge (a \vee a') = (a \wedge a) \vee (a \wedge a') = (a \wedge a) \vee 0 = a \wedge a$ .

**1.25.** Use [Exercise 1.23\(iv\)](#).

**1.26.**  $\mathfrak{m}_x \in \mu(U_f) \Leftrightarrow f(x) \neq 0 \Leftrightarrow f \notin \mathfrak{m}_x \Leftrightarrow \mathfrak{m}_x \in \tilde{U}_f$ .

**1.27.** If  $x \neq y$ , there exists a polynomial  $f \in P(X)$  such that  $f(x) = 0$  but  $f(y) \neq 0$ , so  $\mathfrak{m}_x \neq \mathfrak{m}_y$ , and so  $\mu$  injective. By Nullstellensatz, every maximal ideal in  $P(X)$  is of the form  $\mathfrak{m} = (t_1 - a_1, \dots, t_n - a_n)$ , with  $(a_1, \dots, a_n) \in X$ , so  $\mu$  surjective. By [Exercise 1.26](#),  $\mu$  is a homeomorphism.

**1.28.** Project to coordinate functions.

## 2 Modules

**2.1.** Bézout's identity.

**2.2.** Tensor the exact sequence  $0 \rightarrow \mathfrak{a} \rightarrow A \rightarrow A/\mathfrak{a} \rightarrow 0$ . Calculate the image of  $\mathfrak{a} \otimes M \rightarrow A \otimes M$ .

**2.3.** Consider the residue field  $k$  of  $A$  and pass  $M \otimes_A N$  to vector space  $M_k \otimes_k N_k$ . Use [Exercise 2.2](#) and [Nakayama's lemma](#).

**2.4.** There exist canonical projections.

**2.5.** The definition of flatness only considers the module structure. Use [Exercise 2.4](#).

**2.6.** Construct two homomorphisms.

**2.7.** Proof that  $(A/\mathfrak{p})[x]$  and  $A[x]/\mathfrak{p}[x]$  are isomorphic.

**2.8.** (ii) Use [Proposition 2.15](#).

**2.9.** Pick  $y_1, \dots, y_m \in M$  whose images are the generators of  $M''$ . Use diagram chasing for an element of  $M$ .

**2.10.**  $u(m) + \mathfrak{a}N \in N$ . If  $n \in N$ , we have  $n + \mathfrak{a}N = u(m + \mathfrak{a}M) = u(m) + \mathfrak{a}N$ . Apply [Nakayama's lemma](#) on  $N/u(M)$  where  $\mathfrak{a}(N/u(M)) = (\mathfrak{a}N + u(M))/u(M)$ .

**2.11.** (i),(ii) Consider a residue field and pass to vector spaces. Use the right exactness of tensor product.

(iii) If  $m > n$ , let  $\psi : A^m \xrightarrow{\phi} A^n \hookrightarrow A^m$ . By [Proposition 2.4](#),  $\psi$  satisfies  $\psi^n + a_{n-1}\psi^{n-1} + \cdots + a_0 = 0$ . Pick such equation to have the minimum degree, then  $\psi$  injective implies  $a_0 \neq 0$ . Apply the equation to element  $(0, \dots, 0, 1)$  to obtain contradiction.

**2.12.** Choose  $u_i \in M$  such that  $\phi(u_i)$  are a basis of  $A^n$ , then proof that  $0 \rightarrow \ker(\phi) \rightarrow M \xrightarrow{\phi} A^n \rightarrow 0$  splits.

**2.13.**  $p \circ g = \text{id}_N$  injective  $\Rightarrow g$  injective, so  $0 \rightarrow N \xrightarrow{g} N_B \rightarrow N_B/g(N) \rightarrow 0$  splits.

**2.14.** This question asks to proof  $\mu_i = \mu_j \circ \mu_{ij}$ .

**2.15.** (i) Use [Exercise 2.14](#) to increase indices.

(ii)  $x_i = \sum_k (y_k - \mu_{ki}(y_k))$  is an equation in  $\bigoplus M_i$ , so  $\mu_{ij}$  cannot be applied directly. Use canonical projection  $\pi_i : \bigoplus M_i \rightarrow M_i$  on the equation first.

**2.16** (Universal property of direct limit). Use [Exercise 2.15\(i\)](#) for both existence and uniqueness.

**2.17.** Similar to [Exercise 2.15\(i\)](#), move components into a same index.

**2.18.** Use [Exercise 2.16](#).

**2.19** ( $\varinjlim$  is an exact functor). Diagram chasing. Use [Exercise 2.15](#).

**2.20.** Draw the diagram. Define two homomorphisms by the universal properties of tensor product and direct limit. Verify their compositions are identity maps.

**2.21.** Use [Exercise 2.15\(ii\)](#) on  $\alpha_i(1)$ .

**2.22.** Use [Exercise 2.15](#). Pick a nilpotent or assume  $xy = 0$ .

**2.23.** If  $J \subseteq J'$ ,  $B_J \cong B_J \otimes (A)_{i \in J' \setminus J}$ . Let  $\Gamma$  be the set of all finite subsets of  $\Lambda$ , then  $(B_J)_{J \in \Gamma}$  is the case of [Exercise 2.21](#). We can also use [Exercise 2.17](#) to see its form.

**2.24.** Directly from the definition of Tor functor.

**2.25.** Use [Snake lemma](#).

**2.26.** Let  $M_n$  be generated by  $\{x_1, \dots, x_n\}$ . We have an exact sequence

$$\text{Tor}_1(M_{n-1}, N) \rightarrow \text{Tor}_1(M_n, N) \rightarrow \text{Tor}_1(M_n/M_{n-1}, N).$$

By [Proposition 2.3](#),  $M_n/M_{n-1} \cong A/\mathfrak{a}$ .

Let  $f : \mathfrak{a} \otimes N \rightarrow A \otimes N$ ,  $f(\sum a_i \otimes y_i) = 0$ . Let  $\mathfrak{a}'$  be generated by  $a_i$ . By assumption,  $f|_{\mathfrak{a}' \otimes N}$  is injective, so  $\sum a_i \otimes y_i = 0$  and thus  $\text{Tor}_1(A/\mathfrak{a}, N) = 0$ . Since  $M_0 = A$  and  $\text{Tor}_1(A, N) = 0$ , the exact sequence implies  $\text{Tor}_1(M_n, N) = 0$  for all  $n$ . Then use [Proposition 2.19\(iv\)](#).

Another approach is to use the fact that any module is the direct limit of some finitely generated modules, and direct limit is an exact functor.

**2.27.** (i)  $\Rightarrow$  (ii)  $A/(x)$  is a flat  $A$ -module, then  $\phi : A/(x) \otimes (x) \rightarrow A/(x)$  injective.  $\phi(1 \otimes x) = x = 0$ , so  $A/(x) \otimes (x) = 0$ . Then use [Exercise 2.2](#).

(ii)  $\Rightarrow$  (iii) Let  $x = ax^2$ , then  $ax$  is an idempotent.  $(x) \ni bx = bax^2 = (bx)(ax) \in (ax)$ , so  $(ax) = (x)$ . Similar to [Exercise 1.11\(iii\)](#), every finitely generated ideal is generated by an idempotent  $e$ , so  $A = (e) \oplus (1 - e)$ .

(iii)  $\Rightarrow$  (i)  $N \otimes_A A = (N \otimes_A \mathfrak{a}) \oplus (N \otimes_A \mathfrak{a}')$ , so  $(N \otimes_A \mathfrak{a}) \rightarrow N \otimes_A A$  injective. Use [Exercise 2.26](#).

**2.28.** Use [Exercise 2.27\(ii\)](#).

### 3 Rings and Modules of Fractions

**3.1.** The only if part requires the finitely generated property.

**3.2.** Use [Proposition 1.9](#). Construct the inverse by elementary algebra.

If  $M = \mathfrak{a}M$ ,  $S^{-1}M = S^{-1}\mathfrak{a}$ ,  $S^{-1}M = (S^{-1}\mathfrak{a})(S^{-1}M)$ . Then use [Nakayama's lemma](#) and [Exercise 3.1](#).

**3.3.** Construct the isomorphism by  $(a/s)/(t/1) \mapsto a/(st)$ .

**3.4.** Construct the isomorphism.

**3.5.** If  $a \in A$  is a nilpotent, let  $\mathfrak{p} \supseteq \text{Ann}(a)$ .  $a/1 \in A_{\mathfrak{p}}$  is zero, so  $sa = 0$  for some  $s \in A - \mathfrak{p}$ .

Counter example is  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ .

**3.6.** Use the fact that for any multiplicatively closed set  $S$ , if  $A - S$  is an ideal, it's automatically prime.

**3.7.** (i),(ii) More specifically,  $A - \overline{S} = \bigcup_{\mathfrak{p} \cap S = \emptyset} \mathfrak{p}$ .

(iii)  $\mathfrak{p} \cap (1 + \mathfrak{a}) = \emptyset \Leftrightarrow \mathfrak{a} + \mathfrak{p} \neq (1) \Leftrightarrow \mathfrak{a} + \mathfrak{p} \subseteq \mathfrak{m}$  for some maximal ideal  $\mathfrak{m}$ . Note that the union of a chain of prime ideals is only a maximum ideal.

**3.8.** (v)  $\Rightarrow$  (ii) If  $t/1$  is not a unit in  $S^{-1}A$ , it is contained in a prime ideal  $\mathfrak{q} \subseteq S^{-1}A$ , so  $t \in T \cap \mathfrak{q}^c$ .

**3.9.** Let  $S$  be maximal in [Exercise 3.6](#). If  $S_0 \not\subseteq S$ ,  $S \subsetneq SS_0$ . Then use [Exercise 3.6](#).

(i) Assume that  $S_1$  is larger, then pick a zero-divisor  $x \in S_1$ .

(ii) If  $a/s$  is not a zero-divisor,  $a$  is not a zero-divisor, so  $a \in S_0$ .

(iii) Note that  $s \in S_0$  is a unit.

**3.10.** (i) [Exercise 2.27](#) says for any element  $x \in A$ ,  $x = ax^2$  for some  $a \in A$ .

(ii)  $\Rightarrow$ : Use [Exercise 2.28](#).  $\Leftarrow$ : Consider an  $A_{\mathfrak{m}}$ -module. Use [Proposition 3.10](#).

**3.11.** (i)  $\Rightarrow$  (ii) Use [Exercise 2.27](#) on  $A/\mathfrak{p}$ .

(ii)  $\Rightarrow$  (i) Proof that  $(A/\mathfrak{A})_{\overline{\mathfrak{p}}}$  has no nilpotent, then use [Exercise 1.10](#) and [Exercise 3.10](#).

(ii)  $\Rightarrow$  (iv) Let  $f \in \mathfrak{p}_1$  but  $f \notin \mathfrak{p}_2$ , then  $f/1 \in A_{\mathfrak{p}_1}$  is nilpotent, so  $sf^n = 0$ . We have  $\mathfrak{p}_1 \in X_s$  and  $\mathfrak{p}_2 \in X_{f^n}$ . By [Exercise 1.17](#),  $X_s \cap X_{f^n} = X_{sf^n} = \emptyset$ .

**3.12.** (i)-(iii) Use the torsion element definition.

(iv) Use [Proposition 3.5](#).

**3.13.** Proof they contain in each other. Use [Proposition 3.8](#) for the equivalent conditions.

**3.14.** Pass to  $M/\mathfrak{a}M$ . Use [Proposition 3.8](#).

**3.15.** Let  $\{x_i\}_{i=1}^n$  be generators and  $\{e_i\}_{i=1}^n$  be the canonical basis. Let  $\phi(e_i) = x_i$ , so  $\phi$  is surjective.

Consider a residue field  $k = A/\mathfrak{m}$ , we have  $\text{Tor}_1(k, F) \rightarrow k \otimes \ker \phi \rightarrow k \otimes F \xrightarrow{1 \otimes \phi} k \otimes F \rightarrow 0$ .  $\text{Tor}_1(k, F) = 0$  by balancing Tor theorem;  $1 \otimes \phi$  bijective so  $k \otimes \ker \phi = 0$ . By [Exercise 2.3](#),  $\ker \phi = 0$ .

If  $\{x_i\}_{i=1}^m$  generate  $F$  where  $m < n$ , pick some element  $\{x_i\}_{i=m+1}^n$  so we also have  $\phi(e_i) = x_i$ . By injectiveness of  $\phi$ ,  $\{e_i\}_{i=1}^n$  are linearly dependent.

**3.16.** (i)  $\Rightarrow$  (ii)  $f^*(p^e) = p$ .

(ii)  $\Rightarrow$  (iii) Use [Proposition 1.17](#).

(iii)  $\Rightarrow$  (iv) Every module contains a cyclic submodule, which is isomorphic to  $A/\mathfrak{a}$ . Then use  $B \otimes_A A/\mathfrak{a} \cong B/\mathfrak{a}^e$  and  $\mathfrak{a}^e \subseteq \mathfrak{m}^e$ .

(iv)  $\Rightarrow$  (v) Let the mapping be  $\phi$ . Since  $B$  flat,  $0 \rightarrow (\ker \phi)_B \rightarrow M_B \rightarrow (M_B)_B$  exact. By [Exercise 2.13](#),  $M_B \rightarrow (M_B)_B$  injective, so  $(\ker \phi)_B = 0$ ; by assumption,  $\ker \phi = 0$ .

(v)  $\Rightarrow$  (i) Take  $M = A/\mathfrak{a}$ . Then  $M_B = B/\mathfrak{a}^e$ .



**3.17.** If  $M \rightarrow N$  injective, we have a diagram

$$\begin{array}{ccc} B \otimes_A M & \xrightarrow{\phi} & B \otimes_A N \\ \downarrow g \otimes \text{id}_M & & \downarrow g \otimes \text{id}_N \\ C \otimes_A M & \xrightarrow{\psi} & C \otimes_A N \end{array}$$

$g \circ f$  flat, so  $\psi$  injective.  $g$  faithfully flat, by [Exercise 3.16\(v\)](#),  $B \otimes_A M \rightarrow C \otimes_B (B \otimes_A M)$  injective. By [Proposition 2.15](#),  $C \otimes_B (B \otimes_A M) = C \otimes_A M$ , so  $g \otimes \text{id}_M$  injective.

**3.18.** Use [Exercise 3.3](#) to proof  $B_{\mathfrak{q}} \cong (B_{\mathfrak{p}})_{\mathfrak{q}B_{\mathfrak{p}}}$ . By [Proposition 3.3](#) and [Proposition 3.5](#),  $B_{\mathfrak{q}}$  is flat over  $B_{\mathfrak{p}}$ . Hence  $B_{\mathfrak{q}}$  is flat over  $A_{\mathfrak{p}}$  and satisfies [Exercise 3.16\(iii\)](#).

**3.19.** (v)  $M \cong \bigoplus_{i=1}^n A/\mathfrak{a}_i$ , so  $\text{Supp}(M) = \bigcup_{i=1}^n \text{Supp}(A/\mathfrak{a}_i) = \bigcup_{i=1}^n V(\mathfrak{a}_i) = V(\bigcap_{i=1}^n \mathfrak{a}_i)$ . Proof that  $\bigcap_{i=1}^n \mathfrak{a}_i = \text{Ann}(M)$ .

(vi) Use [Exercise 2.3](#).

(vii) Use [Exercise 2.2](#).

(viii)  $\mathfrak{q} \in f^{*-1}(\text{Supp}(M)) \Leftrightarrow M_{\mathfrak{q}^c} \neq 0 \Leftrightarrow M_{\mathfrak{p}} \neq 0$  and  $B_{\mathfrak{q}} \neq 0 \Leftrightarrow \mathfrak{q} \in \text{Supp}(M) \cap \text{Supp}(B)$ .

**3.20.** Counter example is  $A[x]/(x^2)$ .

**3.21.** (i),(ii) Use [Proposition 3.11\(iv\)](#).

(iii) Use [Exercise 1.21\(iv\)](#).

(iv) To proof  $f^{*-1}(\mathfrak{p}) = \text{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})$ , use  $\mathfrak{q} \in \text{Spec}(S^{-1}B) \Leftrightarrow \mathfrak{q} \cap f(S) = \emptyset$  and  $\mathfrak{q} \in \text{Spec}(C/\mathfrak{p}C) \Leftrightarrow f^{-1}(\mathfrak{q}) \supseteq \mathfrak{p}$ . To proof the equation, note that  $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} = \text{Frac}(A/\mathfrak{p})$ , then apply [Proposition 2.15](#) on  $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \otimes_{A/\mathfrak{p}} A/\mathfrak{p} \otimes_A B$ .

**3.22.** Let  $\mathfrak{p} \in X_f$ , so  $f \notin \mathfrak{p}$ . Then  $\mathfrak{q} \in \text{Spec}(A_{\mathfrak{p}}) \Leftrightarrow \mathfrak{q} \subseteq \mathfrak{p} \Leftrightarrow f \notin \mathfrak{q} \Leftrightarrow \mathfrak{q} \in X_f$ .

**3.23.** (i) Use [Exercise 1.17\(iv\)](#).

(v) Define two homomorphisms by the universal properties of direct limit and local ring.

**3.24.** [Exercise 1.17\(v\)](#) asserts that there exists a finite open cover  $(X_{f_i})$  such that  $\sum_{i=1}^n (f_i) = (f_1, \dots, f_n) = (1)$ , hence  $\sqrt{\sum_{i=1}^n (f_i^N)} = \sqrt{\sum_{i=1}^n \sqrt{(f_i^N)}} = (1)$  and therefore  $\sum_{i=1}^n (f_i^N) = (1)$  for all  $N$ . We can find  $b_i$  such that  $\sum_{i=1}^n b_i f_i^N = 1$ .

Pick  $N$  large enough and let  $s_i = a'_i / f_i^{m_i} = a_i / f_i^N$ . Define  $s = \sum_{i=1}^n a_i b_i$ . Note that  $a_i / f_i^N = a_j / f_j^N$  in  $A(U_i \cap U_j)$ , and use it to verify  $s$  satisfies the condition.

If  $t = s - s'$ ,  $t/1 = 0$  in all  $A_f$ . Pick  $f_i$  are those correspond to the open cover, so  $f_i^M t = 0$  for a large  $M$ . Then  $t = \sum_{i=1}^n b_i f_i^N t = 0$ .

**3.25.** The  $h$  is not a homomorphism. Redefine  $h(x) = f(x) \otimes 1 = 1 \otimes g(x)$ . We have fibers

$$\begin{aligned} f^{*-1}(\mathfrak{p}) &= \text{Spec}(B \otimes_A k(\mathfrak{p})) & g^{*-1}(\mathfrak{p}) &= \text{Spec}(C \otimes_A k(\mathfrak{p})) \\ h^{*-1}(\mathfrak{p}) &= \text{Spec}((B \otimes_A C) \otimes_A k(\mathfrak{p})) \cong \text{Spec}((B \otimes_A k(\mathfrak{p})) \otimes_{k(\mathfrak{p})} (C \otimes_A k(\mathfrak{p}))) \end{aligned}$$

Note that  $B \otimes_A k(\mathfrak{p})$  and  $C \otimes_A k(\mathfrak{p})$  are vector space, so  $\mathfrak{p} \in h^*(T)$  iff  $(B \otimes_A k) \otimes_k (C \otimes_A k) \neq 0$ , iff  $B \otimes_A k \neq 0$  and  $C \otimes_A k \neq 0$ , iff  $\mathfrak{p} \in f^*(Y)$  and  $\mathfrak{p} \in g^*(Z)$ .

**3.26.** The fiber  $f^{*-1}(\mathfrak{p})$  is the spectrum of  $B \otimes_A k(\mathfrak{p})$ . By [Exercise 2.20](#),  $B \otimes_A k(\mathfrak{p}) = \varinjlim (B_{\alpha} \otimes_A k(\mathfrak{p}))$ . By [Exercise 2.21](#),  $f^{*-1}(\mathfrak{p}) = \emptyset \Rightarrow B_{\alpha} \otimes_A k(\mathfrak{p}) = 0$  for some  $\alpha$ , and the reverse is true by the definition of direct limit of rings.

**3.27.**

**3.28.**

**3.29.**

**3.30.**

## 4 Primary Decomposition

4.1. Use [Exercise 1.21\(iv\)](#), so  $\text{Spec}(A/\mathfrak{a}) = V(\mathfrak{a})$ .

4.2. All primary components are prime.

4.3. For any element  $x \in A$ ,  $x(1 - ax) = 0 \in \mathfrak{q}$ .

4.4.  $\mathfrak{m} = \sqrt{\mathfrak{q}} \supsetneq \mathfrak{q} \supsetneq \mathfrak{m}^2$ .

4.5.  $\mathfrak{m}^2$  is  $\mathfrak{m}$ -primary.

4.6. Use Urysohn's lemma to proof that if  $f(x_1) = f(x_2) = 0$  for all  $f \in \mathfrak{q}$ ,  $\mathfrak{q}$  cannot be primary, so there are infinite many minimal prime ideals. Now use [Proposition 1.11\(ii\)](#) to proof that  $(0) = \sqrt{(0)} = \bigcap \mathfrak{p}$  is the intersection of ALL minimal prime ideals.

4.7. (ii),(iv),(v)  $(\mathfrak{a}[x])^c = \mathfrak{a}$ .

(iii) Let  $\bar{f}, \bar{g} \in (A/\mathfrak{q})[x]$ . If  $\bar{f}\bar{g} = 0$  but  $\bar{g} \neq 0$ , by [Exercise 1.2\(iii\)](#),  $\bar{a}\bar{f} = 0$ , so  $aa_i \in \mathfrak{q}$ . Hence  $a_i \in \mathfrak{p}$  and  $a_i^{n_i} \in \mathfrak{q}$ , so  $\bar{a}_i^{n_i} = 0$  and use [Exercise 1.2\(ii\)](#).

4.8.  $k[x_1, \dots, x_n]/\mathfrak{p}_i = k[x_{i+1}, \dots, x_n]$  integer domain, so  $\mathfrak{p}_i$  prime. Note that  $\mathfrak{p}_i^n$  is the set of all homogeneous polynomials in  $k[x_1, \dots, x_i]$  of degree  $\geq n$ .

4.9. (i)  $\Leftarrow$ :  $x$  is contained in all prime ideals containing  $(0 : a)$ , so  $x \in \sqrt{(0 : a)}$ .

(ii) If  $\mathfrak{a} \subseteq \mathfrak{p}$  but  $\mathfrak{a}^e \subseteq S^{-1}\mathfrak{p}' \subsetneq S^{-1}\mathfrak{p}$ , then  $\mathfrak{a} \subseteq \mathfrak{a}^{ec} \subseteq \mathfrak{p}' \subsetneq \mathfrak{p}$ . If  $\mathfrak{b} \subseteq S^{-1}\mathfrak{p}$  but  $\mathfrak{b}^c \subseteq \mathfrak{p}' \subsetneq \mathfrak{p}$ , use  $\mathfrak{b}^{ce} = \mathfrak{b}$  derived from [Proposition 3.11\(i\)](#).  $S^{-1}\mathfrak{p}$  is a prime ideal if  $S$  doesn't meet  $\mathfrak{p}$ .

(iii)  $(0 : a) = \bigcap (\mathfrak{q}_i : a) \subseteq \mathfrak{p}$ . By [Proposition 1.11\(ii\)](#),  $(\mathfrak{q}_i : a) \subseteq \mathfrak{p}$  for some  $i$ . Then use [Proposition 4.4](#) and the minimality of  $\mathfrak{p}$ .

4.10. (iv) Let  $x \neq 0$ ,  $(0 : x) \subseteq \mathfrak{p}$  for some  $\mathfrak{p}$ . If  $x \in S_{\mathfrak{p}}(0)$ , then  $sx = 0$ , so  $s \in (0 : x)$ .

4.11. Note that  $x \in S(\mathfrak{a}) \Leftrightarrow$  there exists an  $s \in S$  such that  $sx \in \mathfrak{a}$ . Show that  $S_{\mathfrak{p}}(0)$  is contained in all  $\mathfrak{p}$ -primary ideals.

If  $0 = \bigcap \mathfrak{q}_i$  is irredundant,  $\sqrt{(0)}$  is the intersection of minimal prime ideals.

4.12. Use [Proposition 3.11\(ii\)](#). Use [Proposition 4.9](#) to show there are at most  $2^n$  different  $S(\mathfrak{a})$ .

4.13. (i) Use [Exercise 4.12](#).

(ii) Use [Proposition 4.8](#) and [Proposition 4.9](#).

(iii) Proof that  $S_{\mathfrak{p}}(\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}) = \mathfrak{p}^{(m+n)}$ .

(iv)  $\Leftarrow$ : Show that  $\mathfrak{p}^{(n)}$  is the smallest  $\mathfrak{p}$ -primary ideal containing  $\mathfrak{p}^n$ .

4.14. Let  $x \notin \mathfrak{q}$ . If  $ab \in (\mathfrak{q} : x)$  but  $a \notin (\mathfrak{q} : x)$ , we have  $xab \in \mathfrak{q}$  but  $xa \notin \mathfrak{q}$ . Thus,  $b \in (\mathfrak{q} : xa) = (\mathfrak{q} : x)$  by maximality, and therefore  $(\mathfrak{q} : x)$  is prime.

4.15. Use [Proposition 4.9](#).

4.16. Use [Proposition 3.11\(iv\)](#) and [Proposition 4.9](#).

4.17.

4.18. (i)  $\Rightarrow$  (L1) By [Proposition 4.9](#),  $S_{\mathfrak{p}}(\mathfrak{a}) = \bigcap \mathfrak{q}_i$ . Pick  $x \in S_{\mathfrak{p}}$ , then  $\mathfrak{q}_i = (\mathfrak{q}_i : x)$  by [Proposition 4.4](#).

(i)  $\Rightarrow$  (L2) Use [Exercise 4.12](#).

4.19. Similar to the proof of [Exercise 4.11](#).

The goal is to proof that there exists  $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$  irredundant, so we need to proof  $\mathfrak{q}_i \not\supseteq \bigcap_{j \neq i} \mathfrak{q}_j$ . Using the fact that  $u\mathfrak{v} \subseteq \mathfrak{q}$  but  $u \not\subseteq \mathfrak{q} \Rightarrow \mathfrak{v} \subseteq \mathfrak{p}$ , we can assume  $\mathfrak{b} = \bigcap_{i=1}^{n-1} \mathfrak{q}_i$  irredundant and  $\mathfrak{p}_n$  maximal among  $\{\mathfrak{p}_i\}$ , so we only need to find  $\mathfrak{q}_n \not\supseteq \mathfrak{b}$ . If such  $\mathfrak{q}_n$  doesn't exist,  $\mathfrak{b} \subseteq S_{\mathfrak{p}_n}(0)$ . Pick a minimal prime ideal  $\mathfrak{p} \subseteq \mathfrak{p}_n$  and use [Exercise 4.10](#) and [Proposition 1.11\(ii\)](#).

4.20.

4.21.

4.22.

4.23.

## 5 Integral Dependence and Valuations

**5.1.**  $f^*(V(I)) = \{f^{-1}(\mathfrak{q}) : \mathfrak{q} \supseteq I\}$ . Use [Theorem 5.10](#) to prove  $f^*(V(I)) \supseteq V(f^{-1}(I))$ .

**5.2.**  $\mathfrak{p} = \ker f$  is prime. By [Theorem 5.10](#), there is a prime ideal  $\mathfrak{q} \subseteq B$ .  $\text{Frac}(B/\mathfrak{q})$  is an algebraic extension of  $\text{Frac}(A/\mathfrak{p})$ , so  $\text{Frac}(A/\mathfrak{p}) \rightarrow \Omega$  can be extended to  $\text{Frac}(B/\mathfrak{q}) \rightarrow \Omega$ .

**5.3.** Use the operator  $- \otimes_A c^n$ .

**5.4.** If  $\left(\frac{1}{x+1}\right)^n + \sum \frac{f_i(x^2-1)}{g_i(x^2-1)(x+1)^i} = 0$ , multiply both side by  $(x+1)^n$  and take  $x = -1$  to get contradiction.

**5.5.** (ii) Use [Theorem 5.10](#) to prove that the contraction of maximal ideals in  $B$  are maximal in  $A$ .

**5.6.** Let  $b_i \in B_i$ ,  $f_i \in A[x_i]$  monic,  $f_i(b_i) = 0$ . Let  $f = (f_1, \dots, f_n)$ , then  $f(b_1, \dots, b_n) = 0$ .

**5.7.** Let  $x \in B$  and  $x$  integral over  $A$ . Let  $x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0$  is the equation of the least degree. Analyze  $x(x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1) = -a_0$ .

**5.8.** Proof that there is a ring  $D \supseteq B$  such that  $f$  and  $g$  factor into linear factors in  $D$ . Use Kronecker's construction.

**5.9.** Let  $f^n + g_{n-1}f^{n-1} + \dots + g_0 = 0$ . Let  $f_1 = f - x^r$  with  $r$  larger than all  $\deg g_i$ , then  $f_1(f_1^{n-1} + h_{n-1}f_1^{n-2} + \dots + h_1) = -h_0$  with  $h_0$  monic. Then use [Exercise 5.8](#).

**5.10.** (a)  $\Rightarrow$  (b) Let  $f^*(V(\mathfrak{q}_1)) = V(I)$ , so  $I \subseteq \mathfrak{p}_1 \subseteq \mathfrak{p}_2$ , and we have  $\mathfrak{p}_2 \in V(I) = f^*(V(\mathfrak{q}_1))$ .

(a')  $\Rightarrow$  (c') By [Exercise 3.23\(v\)](#) and [Exercise 3.26](#),  $f^*(\text{Spec}(B_{\mathfrak{q}})) = f^*(\varinjlim \text{Spec}(B_t)) = \bigcap_t f_t^*(\text{Spec}(B_t))$ . Each  $f_t^*(\text{Spec}(B_t))$  is an open neighborhood of  $\mathfrak{p}$ , so it contains  $A_{\mathfrak{p}}$ .

**5.11.** Use [Exercise 3.18](#) and [Exercise 5.10](#).

**5.12.** Note that  $G$  has identity element so  $x$  is a root of the polynomial  $\prod_{\sigma \in G} (t - \sigma(x))$ . The coefficients are symmetric polynomials of  $\sigma_i$ .  **$G$  acting on  $S^{-1}A$  is defined to be  $\sigma(x/s) = \sigma(x)/\sigma(s)$ .**

**5.13.** Let  $\mathfrak{q} \in P$ . Then  $\sigma(\mathfrak{q}) \cap A^G = \mathfrak{q} \cap A^G = \mathfrak{p}$ , so  $\sigma(\mathfrak{q}) \in P$ . Let  $\mathfrak{q}_1, \mathfrak{q}_2 \in P$  and  $x \in \mathfrak{q}_1$ . Then  $\prod_{\sigma \in G} \sigma(x) \in \mathfrak{q}_1 \cap A^G = \mathfrak{p} \subseteq \mathfrak{q}_2$ , so  $x \in \sigma^{-1}(\mathfrak{q}_2)$ , and so  $\mathfrak{q}_1 \subseteq \bigcup_{\sigma \in G} \sigma(\mathfrak{q}_2)$ . Use [Proposition 1.11\(i\)](#) and [Corollary 5.9](#).

**5.14.** Note that  $\sigma \in G$  fixes  $A$ .

**5.15.**

**5.16.** We have a homomorphism  $f : k[y_1, \dots, y_r] \hookrightarrow k[x_1, \dots, x_n] \twoheadrightarrow A$ . Let  $a = (a_1, \dots, a_r) \in L$  corresponds to a maximal ideal  $\mathfrak{m}_a = (y_1 - a_1, \dots, y_r - a_r)$ . By [Theorem 5.10](#), we can find a prime ideal  $\mathfrak{p}_a \subset A$  such that  $f^{-1}(\mathfrak{p}_a) = \mathfrak{m}_a$ , which is contained in a maximal  $\mathfrak{n}_a$ . Notice that  $\mathfrak{n}_a$  correspond a point in  $X$ .

**5.17.** Consider  $k[t_1, \dots, t_n]/I(X)$ . Use [Exercise 5.16](#).

**5.18 (Zariski's lemma).** Let  $n$  be the number of generators. If  $n = 1$ ,  $x_1$  is invertible, so  $x_1^{-1} = \sum_{i=1}^n c_i x_1^i$ .

If  $n > 1$ , let  $A = k[x_1]$  and  $K_A = \text{Frac}(A)$ . Then  $K_A[x_2, \dots, x_n]$  is a finite algebraic extension of  $K_A$ , so there are monic polynomials  $f_i \in K_A[y]$  such that  $f_i(x_i) = 0$  for  $2 \leq i \leq n$ . Pick  $a \in A$  such that  $af_2 \dots f_n \in A[y]$ , then  $x_2, \dots, x_n$  are integral over  $A_a$ . Therefore,  $K_A \subseteq B$  are integral over  $A_a$ .

If  $x_1$  transcendental over  $k$ ,  $A$  is a UFD. Since every UFD is integrally closed,  $A$  is integrally closed, so  $A_a$  is integrally closed by [Proposition 5.12](#).  $K_A$  integral over  $A_a$ , so  $A_a = K_A$ , but such  $a$  must be a unit in  $A$ .

**5.19.** By [Exercise 5.18](#),  $k[t_1, \dots, t_n]/\mathfrak{m}$  is a finite algebraic extension of  $k$ , so  $k[t_1, \dots, t_n]/\mathfrak{m} \cong k$ . Let  $t_i \mapsto a_i$ , then  $t_i - a_i \in \mathfrak{m}$ . Proof that  $(t_1 - a_1, \dots, t_n - a_n)$  is maximal.

**5.20.** Let  $S = A - \{0\}$ . By [Noether normalization lemma](#),  $S^{-1}B$  is integral over  $S^{-1}A[y_1, \dots, y_r]$  with  $y_1, \dots, y_r \in S^{-1}B$ . The denominators of  $y_1, \dots, y_r$  can be absorbed to  $S^{-1}A$ , so we can assume  $y_1, \dots, y_r \in B$ . Let  $B = A[x_1, \dots, x_n]$ , then  $x_i/1 \in S^{-1}B$  is integral over  $S^{-1}A[y_1, \dots, y_r]$ . Reduce the coefficients of all integral dependence equations to a common denominator  $s$ , we have  $x_i/1$  is integral over  $A_s[y_1, \dots, y_r]$ . Therefore,  $B_s = A_s[x_1/1, \dots, x_n/1]$  is integral over  $A_s[y_1, \dots, y_r]$ .

**5.21.** We have

$$\begin{array}{ccccccc} A & \xrightarrow{\phi_A} & A_s & \longrightarrow & A_s[y_1, \dots, y_n] & \longrightarrow & B_s \xleftarrow{\phi_B} B \\ & & & \searrow f & \searrow f_s & \searrow f'_s & \downarrow g \\ & & & & & & \Omega \end{array}$$

By [Exercise 5.20](#),  $B_s$  integral over  $A_s[y_1, \dots, y_n]$ , so  $g$  is obtained from [Exercise 5.2](#). Then,  $f_s$  by universal property of localization;  $f'_s$  by defining  $y_i \mapsto 0$ .

**5.22.** Let  $v \neq 0$  an element of  $B$ . Then  $B_v$  is integral over  $A$ , so  $f : A \rightarrow \Omega$  can be extended to  $\bar{f} : B_v \rightarrow \Omega$  by [Exercise 5.21](#), with  $s \neq 0$  an element of  $A$  such that  $f(s) \neq 0$ . Find an  $\mathfrak{m}$  such that  $s \notin \mathfrak{m}$  and use [Exercise 5.21](#) again,  $g : A \rightarrow \overline{A/\mathfrak{m}}$  can be extended to  $\bar{g} : B_v \rightarrow \overline{A/\mathfrak{m}}$ . Since  $\bar{g}(s) \neq 0$ ,  $\bar{g}$  is not trivial, so  $\bar{g}(1/v)\bar{g}(v) = \bar{g}(1) = 1$  means  $\bar{g}(v) \neq 0$ .

Since  $\overline{A/\mathfrak{m}}$  integral over  $A/\mathfrak{m}$ ,  $\bar{g}(B_v)$  integral over  $A/\mathfrak{m}$ , so  $\bar{g}(B_v) \cong B_v/\ker(\bar{g})$  is a field and therefore  $\ker \bar{g}$  is maximal. By [Theorem 5.10](#),  $\ker \bar{g} \cap B$  is maximal in  $B$  that doesn't contain  $v$ .

**5.23.** (i)  $\Rightarrow$  (ii) Note that  $f : A \rightarrow A/\ker f$ . Use [Proposition 1.1](#).

(ii)  $\Rightarrow$  (iii) Consider  $A/\mathfrak{p}$ .

(iii)  $\Rightarrow$  (i) If  $\mathfrak{p}$  is not an intersection of maximal ideals, the Jacobson radical  $\mathfrak{J}$  of  $A/\mathfrak{p}$  is not zero. Pick  $f \in \mathfrak{J}$ , then  $(A/\mathfrak{p})_f \neq 0$ . The contraction of a maximal ideal of  $(A/\mathfrak{p})_f$  is  $\mathfrak{q} \in A$  which is not maximal and doesn't contain  $f$ , but all prime ideals strictly containing  $\mathfrak{q}$  contains  $f$ .

**5.24.** (i) Use [Exercise 5.23\(iii\)](#) and [going-up theorem](#).

(ii) Consider  $B/\mathfrak{q}$  and  $A/\mathfrak{q}^c$ . Then use [Exercise 5.22](#), [Exercise 5.23\(i\)](#) and [Proposition 1.1](#).

**5.25.** (i)  $\Rightarrow$  (ii) Similar to the proof of [Exercise 5.22](#), we can find  $\bar{g} : B \rightarrow \overline{A/\mathfrak{m}}$ . Let  $B = A[x_1, \dots, x_n]$ , then  $\bar{g}(x_i)$  integral over  $A/\mathfrak{m}$ , and therefore  $\bar{g}(B)$  is finitely generated as an  $A/\mathfrak{m}$ -algebra. By [Zariski's lemma](#),  $\bar{g}(B)$  is finite over  $A/\mathfrak{m}$ . Note that  $B$  is a field, so  $\bar{g}$  injective.

(ii)  $\Rightarrow$  (i) Take  $B = A/\mathfrak{p}$ . For any  $f \neq 0$  in  $B$ ,  $B_f$  is a finitely generated  $A$ -algebra. If  $B_f$  is a field, it is finite over  $A$ , so integral over  $A$ . The integral dependence equation also asserts that  $B_f$  is integral over  $B$ , so  $B$  is a field by [Proposition 5.7](#). If  $B_f$  is not a field, it has a non-zero prime ideal whose contraction in  $B$  doesn't contain  $f$ . Therefore, the intersection of all prime ideals in  $B$  is zero.

**5.26.** (2)  $\Rightarrow$  (3)  $U_1 \cap X_0 = U_2 \cap X_0 \Rightarrow U_1^c \cap X_0 = U_2^c \cap X_0$ , then take closure.

(i)  $\Rightarrow$  (ii)  $\Rightarrow$  (iii) Consider  $V(\mathfrak{a}) \cap X_f \cap \text{Max}(A)$ . If all maximal ideals in  $V(\mathfrak{a})$  contains  $f$ ,  $V(\mathfrak{a}) \cap X_f = \emptyset$  by [Exercise 5.23\(iii\)](#).

(iii)  $\Rightarrow$  (i) If not Jacobson, there is a prime ideal  $\mathfrak{p}$  such that  $\mathfrak{p} \subsetneq \bigcap_{\mathfrak{q} \supseteq \mathfrak{p}} \mathfrak{q}$ . Let  $f \in \bigcap_{\mathfrak{q} \supseteq \mathfrak{p}} \mathfrak{q} \setminus \mathfrak{p}$ , then  $X_f \cap V(\mathfrak{p}) = \{\mathfrak{p}\}$ .

**5.27.**  $\Rightarrow$ : Use [Theorem 5.21](#) to proof maximal elements are valuation rings.

$\Leftarrow$ : If  $(A, \mathfrak{m}) \subsetneq (A', \mathfrak{m}')$ , pick  $x \in A' \setminus A$ . Then  $x^{-1} \in \mathfrak{m}$  because it is not a unit in  $A$ .

**5.28.** (1)  $\Rightarrow$  (2) Pick  $a \in \mathfrak{a} \setminus \mathfrak{b}$ , and any  $b \in \mathfrak{b}$ . Consider  $a/b$  and  $b/a$ .

(2)  $\Rightarrow$  (1) Consider  $(a) \subseteq (b)$  or  $(b) \subseteq (a)$ , we have  $a = xb$  or  $b = ya$ , so  $y = x^{-1}$ .

**5.29.**  $B$  is a local ring of  $A$  means  $B = A_{\mathfrak{p}}$  for some prime ideal  $\mathfrak{p}$ . Let  $\mathfrak{m}$  be the maximal ideal of  $B$  and let  $\mathfrak{p} = \mathfrak{m} \cap A$ . If  $s \in A$  but  $s \notin \mathfrak{p}$ , then  $s \in B$  and  $s \notin \mathfrak{m}$ , so  $s$  is a unit in  $B$ . Let  $A_{\mathfrak{p}} \rightarrow B$  to be  $x/s \mapsto s^{-1}x$ , which is injective;  $\mathfrak{p}A_{\mathfrak{p}} \subseteq B$ , so  $B$  dominates  $A_{\mathfrak{p}}$ . By [Exercise 5.28](#),  $A_{\mathfrak{p}}$  is a valuation ring, then use [Exercise 5.27](#).

**5.30.**  $xy^{-1} \in A$  or  $x^{-1}y \in A$ , so  $\geq$  is a total order. Also  $(xz)(yz)^{-1} = xy^{-1}$ .

**Treat  $v$  as a representation map.** If  $v(x) \geq v(y)$ ,  $(x+y)y^{-1} = xy^{-1} + 1 \in A$ .

**5.31.** Use the given conditions to verify the set is closed under addition and multiplication. Then verify it is a valuation ring by taking  $x = y = 1$  and  $y = x^{-1}$  in the first condition.

**5.32.**

**5.33.**

**5.34.**

**5.35.**

## 6 Chain Conditions

**6.1.** (i) Let  $x \in \ker(u)$ . Since  $u^n$  surjective, there is a  $y \in M$  such that  $u^n(y) = x$ . But  $u^{n+1}(y) = u(x) = 0$ , means that  $y \in \ker(u^{n+1}) = \ker(u^n)$ .

(ii) Let  $x \in M$ . Then  $u^n(x) \in \operatorname{im}(u^n) = \operatorname{im}(u^{n+1})$ , so there is a  $y \in M$  such that  $u^{n+1}(y) = u^n(x)$ . But  $u^n$  injective, means that  $u(y) = x$ .

**6.2.** If  $M$  is not Noetherian, there is a submodule  $N$  which is not finitely generated. Construct a chain of finitely generated submodules by the generators of  $N$ .

**6.3.** Consider  $0 \rightarrow N_2/(N_1 \cap N_2) \rightarrow M/(N_1 \cap N_2) \rightarrow M/N_2 \rightarrow 0$ . Use [Proposition 2.1](#).

**6.4.** Let  $m_1, \dots, m_n$  be generators of  $M$ , then  $\mathfrak{a} = \bigcap \operatorname{Ann}(m_i)$ .  $A/\operatorname{Ann}(m_i) \cong Am_i \subseteq M$ , then use [Exercise 6.3](#).

**6.5.** Use the open set definition.

**6.6.** (i)  $\Rightarrow$  (iii) Use [Exercise 6.5](#).

(ii)  $\Rightarrow$  (i) Consider a chain of ascending open sets in  $X$ .

**6.7.** Let  $\Sigma$  be the collection of closed subsets of  $X$  which are not finite union of irreducible closed subspaces. Then  $\Sigma$  has minimal element  $X_0$ . Proof that  $X_0$  cannot be either irreducible or reducible.

**6.8.** The converse isn't true. The counter example is  $k[x_1, \dots]/(x_1^2, \dots)$ .

**6.9.** The irreducible components of  $\operatorname{Spec}(A)$  is  $V(\mathfrak{p})$  where  $\mathfrak{p}$  minimal ([Exercise 1.20\(iv\)](#)).

**6.10.** Use [Exercise 3.19\(v\)](#).

**6.11.** By [Exercise 6.7](#),  $V(\mathfrak{b})$  is a finite union of  $V(\mathfrak{q}_i)$ , where  $\mathfrak{q}_i$  minimal and containing  $\mathfrak{b}$ . Then use [Exercise 5.10\(c\)](#).

**6.12.** The converse isn't true. A counter example is an infinite product  $\prod \mathbb{Z}/2\mathbb{Z}$ .

## 7 Noetherian Rings

**7.1** (Cohen's theorem). Let  $\mathfrak{a}$  be a maximal element in  $\Sigma$ . If  $xy \in \mathfrak{a}$  but  $x, y \notin \mathfrak{a}$ , then  $\mathfrak{a} + (x)$  is finitely generated. Let  $\{x, a_1, \dots, a_n\}$  be its generators and let  $\mathfrak{a}_0 = (a_1, \dots, a_n)$ . Proof that  $\mathfrak{a}_0 \not\subseteq \mathfrak{a}$  is impossible. Therefore, if  $u \in \mathfrak{a} \subseteq \mathfrak{a} + (x) = \mathfrak{a}_0 + (x)$ ,  $u = a_0 + bx$  with  $b \in \mathfrak{a}$ , so  $\mathfrak{a} \subseteq \mathfrak{a}_0 + x \cdot (\mathfrak{a} : x)$ ; we also have  $x \cdot (\mathfrak{a} : x) \subseteq \mathfrak{a}$ . Note that  $y \in (\mathfrak{a} : x)$ .

**7.2.**  $\Rightarrow$ : [Exercise 1.5\(ii\)](#).

$\Leftarrow$ :  $(a_0, a_1, \dots) \subseteq \mathfrak{R}$  and use [Corollary 7.15](#).

**7.3.** (i)  $\Rightarrow$  (ii) Use [Proposition 4.8](#) and [Proposition 4.4](#).

(ii)  $\Rightarrow$  (iii) Let  $S = \{1, x, x^2, \dots\}$ . Proof that  $\bigcup (\mathfrak{a} : x^n) = S(\mathfrak{a})$ .

(iii)  $\Rightarrow$  (i) Similar to the proof of [Proposition 7.12](#).

**7.4.** (i)  $\mathfrak{p} = \{(z - c) : |c| = 1\}$  is prime, and the ring is  $(\mathbb{C}[x])_{\mathfrak{p}}$ . Use [Proposition 7.3](#).

- (ii) Use [Exercise 1.5\(i\)](#). If  $f = z^n g$  with  $g$  invertible,  $(f) = (z^n)$ , so the only ideals are  $(0)$  and  $(z^n)$ .  
 (iii)  $\cdots \subsetneq (\sin \pi n z) \subsetneq (\sin \pi(n+1)z) \subsetneq \cdots$  is an ascending chain.  
 (iv) The ring is  $\mathbb{C}[z^{k+1}]$ .  
 (v) The ring is  $\mathbb{C}[z, z^2, \dots, wz, wz^2, \dots, w^2 z, w^2 z^2, \dots]$ .

**7.5.** Use [Exercise 5.12](#) and [Proposition 7.8](#).

**7.6.** A ring is finitely generated if it is finitely generated as a  $\mathbb{Z}$ -algebra.  $\mathbb{Z}$  is a Jacobson ring. By [Exercise 5.25](#),  $K$  is finite over  $\mathbb{Z}$ , so it is of the form  $\mathbb{Z}^n \oplus \bigoplus_i \mathbb{Z}/p_i \mathbb{Z}$  due to the addition operation forms a finitely generated abelian group. If  $\text{char } K = 0$ ,  $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{Z}^n$ , then use [Proposition 7.8](#).

**7.7.** Use ascending chain condition.

**7.8.**  $A \cong A[x]/(x)$ .

**7.9.** Let  $\mathfrak{a}_0 \subseteq \mathfrak{a}_1 \subseteq \cdots$  be a chain in  $A$ . Since  $\mathfrak{a}_0$  is contained in only finite many maximal ideals, there is a large enough  $n$  such that  $S_{\mathfrak{m}}^{-1} \mathfrak{a}_n = S_{\mathfrak{m}}^{-1} \mathfrak{a}_{n+1} = \cdots$  for all  $\mathfrak{m} \supseteq \mathfrak{a}_0$ . We also have  $S_{\mathfrak{m}}^{-1} \mathfrak{a}_i = A_{\mathfrak{m}}$  for all  $i$ , when  $\mathfrak{m} \not\supseteq \mathfrak{a}_0$ . Then apply [Proposition 3.8](#) on  $S_{\mathfrak{m}}^{-1}(\mathfrak{a}_{n+i}/\mathfrak{a}_n)$ .

**7.10.**  $M[x] = M \otimes A[x]$ , so it is a homomorphic image of  $M \times A[x]$ . Use [Proposition 7.1](#).

**7.11.** Counter example is an infinite product  $\prod \mathbb{Z}/2\mathbb{Z}$ .

**7.12.** By [Exercise 3.16\(i\)](#),  $\mathfrak{a} \mapsto \mathfrak{a}^e$  is injective. Then use ascending chain condition.

**7.13.** Transfer the Noetherian property  $B \rightarrow B_{\mathfrak{p}} \rightarrow B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$  and use [Exercise 6.8](#).

**7.14** ([Hilbert's Nullstellensatz](#)). If  $f \notin \sqrt{\mathfrak{a}}$ , there is a  $\mathfrak{p} \supseteq \mathfrak{a}$  such that  $f \notin \mathfrak{p}$ .  $(A/\mathfrak{p})_{\overline{f}}$  is a finitely generated  $k$ -algebra, so by [Zariski's lemma](#),  $\left((A/\mathfrak{p})_{\overline{f}}\right)/\mathfrak{m} = k$ . Let  $x_i$  be the images of  $t_i$  in  $\left((A/\mathfrak{p})_{\overline{f}}\right)/\mathfrak{m}$ , then  $x = (x_1, \dots, x_n)$  satisfies  $x \in V$  but  $f(x) \neq 0$ .

**Rabinowitsch trick:** Let  $f_1, \dots, f_m$  be generators of  $\mathfrak{a}$ . If  $f \in I(V)$ , then  $f_1, \dots, f_m, 1 - t_0 f$  has no zeros in  $k[t_0, t_1, \dots, t_n]$ . Apply [Exercise 5.17](#),  $1 = g_0(1 - t_0 f) + \sum_{i=1}^m g_i f_i$ . Now put  $t_0 = 1/f$ , we have  $g_i \in k[t_1, \dots, t_n, 1/f]$ . Therefore,  $g_i = h_i/f^{N_i}$  with  $h_i \in k[t_1, \dots, t_n]$ . Hence we have  $f^N = \sum_{i=1}^m h_i f^{N-N_i} f_i$  for all  $N \geq \max_i \{N_i\}$ .

**7.15.** (iv)  $\Rightarrow$  (i) Let  $x_1, \dots, x_n$  be elements of  $M$  whose images in  $M/\mathfrak{m}M$  form a basis of  $k$ -vector space. Let  $N = (x_1, \dots, x_n)$ , so  $N + \mathfrak{m}M = M$ . Since  $\mathfrak{J} = \mathfrak{m}$ , apply [Nakayama's lemma](#) on  $M/N$ , we have  $M = N$ . Let  $F$  be a free  $A$ -module with basis  $e_1, \dots, e_n$ , and  $\phi: F \rightarrow M$  be  $\phi(e_i) = x_i$ . Then

$$\text{Tor}_1^A(k, M) \rightarrow k \otimes_A \ker \phi \rightarrow k \otimes_A F \xrightarrow{1 \otimes \phi} k \otimes_A M \rightarrow 0$$

exact. Since  $k \otimes F$  and  $k \otimes M = M/\mathfrak{m}M$  are vector spaces of the same dimension,  $1 \otimes \phi$  is an isomorphism, so  $k \otimes \ker \phi = 0$ . By [Exercise 2.3](#),  $\ker \phi = 0$ .

**7.16.** Use [Proposition 3.10](#) and [Exercise 7.15](#).

**7.17.**

**7.18.**

**7.19.**

**7.20.** (i)  $\Rightarrow$ : We have a chain  $\mathcal{F}_0 \subseteq \mathcal{C}_1 \subseteq \mathcal{F}_1 \subseteq \mathcal{C}_2 \subseteq \cdots \subseteq \mathcal{F}$ , where  $\mathcal{F}_0$  is the topology,  $\mathcal{C}_i = \mathcal{F}_{i-1} \cup \{S^c : S \in \mathcal{F}_{i-1}\}$ , and  $\mathcal{F}_i$  is a collection of all finite intersections of sets in  $\mathcal{C}_i$ .

(ii)  $\Rightarrow$ : Use the open set definition.

**7.21.**  $\Rightarrow$ : Use [Exercise 7.20](#).

$\Leftarrow$ : If  $E \notin \mathcal{F}$ , let  $\mathcal{C} = \{X' \subseteq X : X' \text{ closed, } E \cap X' \notin \mathcal{F}\}$ . By Noetherian property,  $\mathcal{C}$  has a minimal element  $X_0$ . Use the closed set definition to show  $X_0$  is irreducible. If  $\overline{E \cap X_0} \neq X_0$ ,  $E \cap \overline{E \cap X_0} \in \mathcal{F}$  by minimality, but  $E \cap \overline{E \cap X_0} = E \cap X_0$ . If  $U_0 \subseteq E \cap X_0$ , let  $U_0 = U \cap X_0$  where  $U$  is open in  $X$ , then  $E \cap X_0 = U_0 \cup (E \cap U^c \cap X_0)$ .

**7.22.**  $\Leftarrow$ : If  $E$  is not open, let  $\mathcal{C} = \{X' \subseteq X : X' \text{ closed, } E \cap X' \text{ is not open}\}$  and use the notations similar to [Exercise 7.21](#). If  $U_0 \subseteq E \cap X_0$ ,  $U^c \cap X_0$  is closed, so by minimality  $E \cap U^c \cap X_0$  is open.

**7.23.** Hint.

**7.24.** Hint.

**7.25.** Use [Exercise 5.11](#) and [Exercise 7.24](#).

**7.26.**

**7.27.**

## 8 Artinian Rings

**8.1.** Use [Corollary 7.16](#) on local ring  $A_{\mathfrak{p}_i}$ . Then apply the contraction. If  $\mathfrak{q}_i$  is isolated, then  $\mathfrak{p}_i$  is minimal.

**8.2.** (i)  $\Rightarrow$  (ii) Use [Proposition 8.1](#).

(iii)  $\Rightarrow$  (i) Use [Proposition 8.5](#).

**8.3.** (i)  $\Rightarrow$  (ii) By [Proposition 8.7](#), we can assume that  $A$  is an Artinian local ring. By [Proposition 8.4](#),  $\mathfrak{m}^n = 0$  for some  $n$ , so we have a chain  $A = \mathfrak{m}^0 \supseteq \mathfrak{m}^1 \supseteq \cdots \supseteq \mathfrak{m}^n = 0$ .

By [Zariski's lemma](#),  $A/\mathfrak{m}$  is a finitely generated  $k$ -module. By [Proposition 8.5](#),  $A$  is Noetherian, so  $\mathfrak{m}^i$  is a finitely generated  $A$ -module, and therefore  $\mathfrak{m}^i/\mathfrak{m}^{i+1}$  is a finitely generated  $A/\mathfrak{m}$ -module. Use

$$\begin{aligned} \dim_k(\mathfrak{m}^i/\mathfrak{m}^{i+1}) &= \dim_{A/\mathfrak{m}}(\mathfrak{m}^i/\mathfrak{m}^{i+1}) \dim_k(A/\mathfrak{m}); \\ \dim_k(\mathfrak{m}^i) &= \dim_k(\mathfrak{m}^{i+1}) + \dim_k(\mathfrak{m}^i/\mathfrak{m}^{i+1}), \end{aligned} \quad \text{where } 0 \rightarrow \mathfrak{m}^{i+1} \rightarrow \mathfrak{m}^i \rightarrow \mathfrak{m}^i/\mathfrak{m}^{i+1} \rightarrow 0$$

to finish the proof.

(ii)  $\Rightarrow$  (i)  $A$  is a  $k$ -vector space of finite dimension. Use [Proposition 6.10](#).

**8.4.** (i)  $\Rightarrow$  (iii) The finitely-generated property is preserved for localization and quotient of modules.

(ii)  $\Rightarrow$  (iii)  $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$  is a finitely generated  $k(\mathfrak{p})$ -algebra. By [Hilbert's basis theorem](#), any finitely generated  $k(\mathfrak{p})$ -algebra is Noetherian. Then use [Exercise 8.2](#) and [Exercise 8.3](#).

(iii)  $\Rightarrow$  (ii),(iv)  $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$  is Noetherian and integral over the field  $k(\mathfrak{p})$ . If  $\mathfrak{q} \in B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$  is prime,  $(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})/\mathfrak{q}$  is an integral domain, so it is a field by [Proposition 5.7](#). Hence, every prime ideal in  $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$  is maximal, which means  $\dim(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}) = 0$ . Thus,  $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$  is Artinian by [Proposition 8.5](#).

Then use [Proposition 8.3](#).

A counter example is  $f : \mathbb{Q} \hookrightarrow \overline{\mathbb{Q}}$  where  $\overline{\mathbb{Q}}$  is an algebraic closure of  $\mathbb{Q}$ .

**8.5.** Use [Noether normalization lemma](#) and let  $\mathfrak{m}$  be a maximal ideal in  $k[y_1, \dots, y_r]$ . By [Exercise 8.4](#),  $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$  is finite over  $k[y_1, \dots, y_r]/\mathfrak{m} = k$ , so by [Exercise 8.3](#) and [Proposition 8.3](#),  $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$  only has finite many maximal ideals, correspond to finite many points in  $X$ .

Let  $N$  be the number of generators of  $A$  over  $k[y_1, \dots, y_r]$ . The  $k$ -dimension of  $k$ -vector space  $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$  is uniformly bounded by  $N$ . If  $\mathfrak{n}_1, \dots, \mathfrak{n}_M$  are maximal ideals in  $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$ , by [Proposition 1.10](#),  $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}} \rightarrow \prod_{i=1}^M (A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}})/\mathfrak{n}_i$  is surjective, so  $\dim_k(A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}) \geq \sum_{i=1}^M \dim_k((A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}})/\mathfrak{n}_i) \geq M$ .

**8.6.** Consider  $(A/\mathfrak{q})_{\overline{\mathfrak{p}}}$ , which is an Artinian local ring. Use the following claims then [Proposition 6.7](#).

- If  $A$  is an Artinian local ring,  $\mathfrak{m}$  its only maximal ideal. Then every ideal in  $A$  is  $\mathfrak{m}$ -primary. Radical is the intersection of the only prime ideal.
- There is a 1-1 correspondence between  $\mathfrak{p}$ -primary ideals in  $A$  and  $S^{-1}\mathfrak{p}$ -primary ideals in  $S^{-1}A$ . Use [Proposition 3.11\(i\)](#) and [Proposition 4.9](#).



## 9 Discrete Valuation Rings and Dedekind Domain

**9.1.** [Proposition 3.11\(iv\)](#) asserts that  $\dim S^{-1}A$  can only be 0 and 1. If  $\dim S^{-1}A = 0$ , it is  $\text{Frac } A$ ; if  $\dim S^{-1}A = 1$ , use [Proposition 5.12](#).

By [Corollary 3.15](#),  $S^{-1}(M^{-1}) = S^{-1}(A : M) = (S^{-1}A : S^{-1}M) = (S^{-1}M)^{-1}$ . We also have  $S^{-1}(MN) = (S^{-1}M)(S^{-1}N)$ , so the map  $M \mapsto S^{-1}M$  is a homomorphism. Use [Proposition 3.11\(i\)](#). Note that  $(u) \mapsto S^{-1}(u) = (S^{-1}A)u$  maps  $P$  into  $P'$ , so  $H \rightarrow H'$  is surjective.

**9.2.** Localize and use [Proposition 3.8](#) to the final move, we can assume that  $A$  is a DVR.

[Proposition 9.2\(iii\)\(v\)](#) assert that  $c(f) = (x^n)$  where  $x$  is the generator of  $\mathfrak{m}$ , so  $f = x^n f_0$  where  $f_0$  is primitive. Apply [Exercise 1.2\(iv\)](#) to  $f_0, g_0$  and note that for any  $a \in A$ ,  $ac(f) = c(af)$ .

**9.3.**  $\Rightarrow$ :  $(x_1, x_2) = (x_1)$  if  $x_1 x_2^{-1} \in A$ , so every ideal is principal. Then use [Proposition 9.2\(iii\)](#).  
 $\Leftarrow$ : If  $v(y) \geq v(x)$ ,  $y = (yx^{-1})x$ , so every ideal in DVR is of the form  $\mathfrak{m}_k = \{z \in A : v(z) \geq k\}$ .

**9.4.** For any non-zero  $x \in A$ ,  $x \in \mathfrak{m}^n$  but  $x \notin \mathfrak{m}^{n+1}$  for some  $n$ . Define a discrete valuation  $v(x) = n$ .

**9.5.** By [Exercise 3.13](#) and [Exercise 7.16](#), and notice that torsion free module over a PID is flat.

**9.6.** From [Theorem 9.3\(iii\)](#), [Proposition 9.2\(v\)](#), and the structure theorem for finitely generated modules over a PID,  $M_{\mathfrak{p}} = \bigoplus_{i=1}^N A_{\mathfrak{p}}/\mathfrak{p}^{n_i} A_{\mathfrak{p}} = \bigoplus_{i=1}^N (A/\mathfrak{p}^{n_i})_{\mathfrak{p}}$ . Define a homomorphism  $\phi : M \rightarrow \bigoplus_{\mathfrak{p}} M_{\mathfrak{p}}$  by  $m \mapsto (m/1, \dots)$ , so each  $\phi_{\mathfrak{p}}$  is an isomorphism. By [Proposition 3.9](#),  $\phi$  is an isomorphism.

By [Exercise 3.19\(v\)](#),  $\text{Supp}(M) = V(\text{Ann}(M))$ . From [Proposition 9.1](#) and [Theorem 9.3\(ii\)](#),  $\text{Ann}(M) = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_k^{n_k}$ . Use  $\dim A = 1$  to proof that  $V(\text{Ann}(M))$  is finite.

**9.7.** From [Proposition 9.1](#) and [Theorem 9.3\(ii\)](#),  $\mathfrak{a} = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_k^{n_k}$ . By [Proposition 1.10](#),  $A \rightarrow \prod_{i=1}^k A/\mathfrak{p}_i^{n_i}$  is a surjective with kernel  $\mathfrak{a}$ . Note that each component  $A/\mathfrak{p}_i^{n_i}$  is local so is PID.

If  $a \in \mathfrak{b}$ ,  $\mathfrak{b}/(a) = (\bar{b})$  and therefore  $\mathfrak{b} = (a, b)$ .

**9.8.** Localize and note that all ideals in DVR are  $\{z \in A : v(z) \geq k\}$ . Use [Proposition 3.8](#) to pull back.

**9.9.** This is equivalent to say that

$$A \xrightarrow{\phi} \bigoplus_{i=1}^n A/\mathfrak{a}_i \xrightarrow{\psi} \bigoplus_{i < j} A/(\mathfrak{a}_i + \mathfrak{a}_j)$$

is exact, where  $\phi(x) = (x + \mathfrak{a}_i)_{i=1}^n$ , and  $\psi(x_i + \mathfrak{a}_i)_{i=1}^n = (x_i - x_j + \mathfrak{a}_i + \mathfrak{a}_j)_{i < j}$ . If  $A$  is DVR, we have  $\mathfrak{a}_i + \mathfrak{a}_j = \mathfrak{a}_{\min(i,j)}$ . If  $(x_i + \mathfrak{a}_i)_{i=1}^n \in \ker \psi$ ,  $x_i + \mathfrak{a}_i = x_j + \mathfrak{a}_i$  for all  $i < j$ , so take  $j = n$  for all  $i$  and we have  $\phi(x_n) \in \ker \psi$ . Then use [Theorem 9.3\(iii\)](#) and [Proposition 3.8](#) for Dedekind domain.

## 10 Completion

**10.1.** Some completions are calculated as follows

- $A/p^n A = \bigoplus_{k \geq 0} \mathbb{Z}/p\mathbb{Z}$ , where the transition maps are all identity.
- $A_n = \alpha^{-1}(p^n B) = \bigoplus_{k > n} \mathbb{Z}/p\mathbb{Z}$ , so  $A/A_n = \bigoplus_{k=0}^n \mathbb{Z}/p\mathbb{Z}$ . The transition maps mean, if  $(x^{(1)}, x^{(2)}, \dots) \in \varprojlim A/A_n$ , then  $x_i^{(i)} = x_i^{(i+1)} = \dots$  for all  $i$ .
- $B/p^n B = (\bigoplus_{k=0}^n \mathbb{Z}/p^k \mathbb{Z}) \oplus (\bigoplus_{k > n} \mathbb{Z}/p^n \mathbb{Z})$ . The transition maps are similar to  $A/A_n$ .

The sequence  $0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{p} pB \rightarrow 0$  is exact. Since  $p$ -adic topology and induced topology on  $pB$  coincide, by [Corollary 10.3](#),  $\ker(\hat{p}) = \varprojlim A/A_n$ .

**10.2.** The explicit form of the given exact sequence is

$$0 \rightarrow \bigoplus_{k > n} \mathbb{Z}/p\mathbb{Z} \rightarrow \bigoplus_{k \geq 0} \mathbb{Z}/p\mathbb{Z} \rightarrow \bigoplus_{k=0}^n \mathbb{Z}/p\mathbb{Z} \rightarrow 0.$$

Use [Exercise 10.1](#), and note that  $\varprojlim \bigoplus_{k > n} \mathbb{Z}/p\mathbb{Z} = 0$ . The definition of  $\varprojlim^1 A$  in this book is [here](#).



**10.3.** Denote  $N = \bigcap_{\mathfrak{m} \supseteq \mathfrak{a}} \ker(M \rightarrow M_{\mathfrak{m}})$ . Let  $x \in \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$ . By [Proposition 10.17](#), we have  $(1+a)x = 0$ . If  $\mathfrak{m} \supseteq \mathfrak{a}$ ,  $1+a \in A - \mathfrak{m}$ , so  $x \in N$ . Since  $(\ker(M \rightarrow M_{\mathfrak{m}}))_{\mathfrak{m}} = 0$ , we have  $N_{\mathfrak{m}} = 0$  for all  $\mathfrak{m} \supseteq \mathfrak{a}$ . By [Exercise 3.14](#),  $N = \mathfrak{a}N$ , and therefore  $N = \mathfrak{a}N = \mathfrak{a}^2 N = \dots$ . Thus,  $N = \bigcap_{n=1}^{\infty} \mathfrak{a}^n N \subseteq \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$ .

We can see  $N$  is also equal to  $\bigcap_{\mathfrak{p} \supseteq \mathfrak{a}} \ker(M \rightarrow M_{\mathfrak{p}})$ . Therefore,  $\text{Supp}(M) \cap V(\mathfrak{a}) = \emptyset$  iff  $N = M$ , iff  $M = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$ , iff  $M = \mathfrak{a}M$ . By [Exercise 2.2](#), [Proposition 10.13](#) and [Proposition 10.15\(iii\)](#),  $\hat{M}/\hat{\mathfrak{a}}\hat{M} = \hat{A}/\hat{\mathfrak{a}}\hat{A} \otimes_{\hat{A}} \hat{M} = A/\mathfrak{a} \otimes_A (M \otimes_A \hat{A}) = M/\mathfrak{a}M \otimes_A \hat{A}$ . Since  $\hat{A} \neq 0$ , by [Exercise 2.3](#),  $M/\mathfrak{a}M = 0$  iff  $\hat{M}/\hat{\mathfrak{a}}\hat{M} = 0$ . By [Proposition 10.15\(iv\)](#) and [Nakayama's lemma](#),  $\hat{M} = \hat{\mathfrak{a}}\hat{M}$  iff  $\hat{M} = 0$ .

**10.4.**  $x$  not a zero-divisor, so  $A \xrightarrow{x} A$  is injective. Use [Proposition 10.14](#) and apply the functor  $\hat{A} \otimes_A -$ .

Counter example is  $k[x, y]/f$ , where  $f = y^2 - x^2 - x^3$ . The completion is  $k[[x, y]]/f$ .  $f$  is irreducible in  $k[x, y]$ , but  $f = (y - x\sqrt{1+x})(y + x\sqrt{1+x})$  is 0 in  $k[[x, y]]/f$ .

**10.5.** We can deduce from [Proposition 10.2](#) that taking  $\mathfrak{a}$ -adic completion on  $0 \rightarrow b^m M \rightarrow M \rightarrow M/b^m M \rightarrow 0$  preserves exactness. Thus,  $(M/b^m M)^{\mathfrak{a}} \cong M^{\mathfrak{a}}/b^m M^{\mathfrak{a}}$ , and therefore

$$(M^{\mathfrak{a}})^{\mathfrak{b}} = \varprojlim_m \left( \varprojlim_n (M/\mathfrak{a}^n M)/(\mathfrak{b}^m (M/\mathfrak{a}^n M)) \right) = \varprojlim_m \left( \varprojlim_n M/(\mathfrak{a}^n + \mathfrak{b}^m) M \right).$$

On the other hand, we have  $(\mathfrak{a} + \mathfrak{b})^{2n} \subseteq \mathfrak{a}^n + \mathfrak{b}^n \subseteq (\mathfrak{a} + \mathfrak{b})^n$ . If  $x \in (\mathfrak{a} + \mathfrak{b})^n$ ,  $x + \mathfrak{a}^n + \mathfrak{b}^n \subseteq (\mathfrak{a} + \mathfrak{b})^n$ , which means each set in the basis  $\{(\mathfrak{a} + \mathfrak{b})^n\}$  is open in the topology generated by the basis  $\{\mathfrak{a}^n + \mathfrak{b}^n\}$ , and vice versa. Thus, they generate a same topology, and therefore

$$M^{\mathfrak{a}+\mathfrak{b}} = \varprojlim M/(\mathfrak{a} + \mathfrak{b})^n M = \varprojlim M/(\mathfrak{a}^n + \mathfrak{b}^n) M.$$

We construct two homomorphisms  $\varprojlim_m \left( \varprojlim_n M/(\mathfrak{a}^n + \mathfrak{b}^m) M \right) \xrightarrow[\Psi]{\Phi} \varprojlim M/(\mathfrak{a}^n + \mathfrak{b}^n) M$ . When  $k \geq \max(m, n)$ ,  $\mathfrak{a}^k + \mathfrak{b}^k \subseteq \mathfrak{a}^n + \mathfrak{b}^m$ , so there is a canonical quotient map  $\psi_{nm} : M/(\mathfrak{a}^k + \mathfrak{b}^k) M \rightarrow M/(\mathfrak{a}^n + \mathfrak{b}^m) M$ . Let  $\Phi((x)_{nm}) = (x)_{kk}$  be the diagonal map, and let  $\Psi((y)_k) = \psi_{nm}((y)_k)$ .

**10.6.** If  $\mathfrak{a} \subseteq \mathfrak{J}$ , let  $x \notin \mathfrak{m}$ . We can find an open neighborhood  $x + \mathfrak{a}$  of  $x$  such that  $(x + \mathfrak{a}) \cap \mathfrak{m} = \emptyset$ .

By [Proposition 10.1](#), the closure of  $\mathfrak{m}$  is  $\bigcap_{n \geq 1} (\mathfrak{m} + \mathfrak{a}^n)$ . If  $\mathfrak{a} \not\subseteq \mathfrak{m}$ , we have  $\mathfrak{m} + \mathfrak{a}^n = A$ , otherwise  $\mathfrak{a} \subseteq \sqrt{\mathfrak{a}^n} \subseteq \mathfrak{m}$ .

**10.7.** Let  $M$  be a finitely generated  $A$ -module. By [Proposition 10.13](#),  $\hat{M} = M_{\hat{A}}$ , so  $\ker(M \rightarrow M_{\hat{A}}) = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$ .

If  $\mathfrak{a} \subseteq \mathfrak{J}$ , every element in  $1 + \mathfrak{a}$  is a unit, and by [Proposition 10.17](#),  $\bigcap_{n=1}^{\infty} \mathfrak{a}^n M = 0$ . If  $N$  is an arbitrary  $A$ -module and  $x \in \ker(N \rightarrow N_{\hat{A}})$ , then  $x \in \ker(Ax \rightarrow (Ax)_{\hat{A}})$  and  $Ax$  is finitely generated.

If  $\mathfrak{a} \not\subseteq \mathfrak{m}$ , we have  $\mathfrak{m} + \mathfrak{a}^n = A$ .  $(A/\mathfrak{m})/(\mathfrak{a}^n(A/\mathfrak{m})) = A/(\mathfrak{m} + \mathfrak{a}^n) = 0$ , so the  $\mathfrak{a}$ -adic completion of  $A/\mathfrak{m}$  is 0. Hence,  $A/\mathfrak{m} \rightarrow (A/\mathfrak{m})_{\hat{A}}$  isn't injective.

**10.8.** By [Exercise 1.5\(i\)](#) and [Corollary 1.6\(i\)](#), the only maximal ideal of  $B$  is  $\mathfrak{m} = (z_1, \dots, z_n)$ .

The  $\mathfrak{m}$ -adic topologies on  $A$ ,  $B$  and  $C$  are respectively equal to the induced topologies from the  $\mathfrak{m}$ -adic topology on  $\mathbb{C}[z_1, \dots, z_n]$ . Apply [Proposition 10.2](#) to  $\mathbb{C}[z_1, \dots, z_n] \subseteq A \subseteq B \subseteq C$ .

By [Proposition 10.14](#),  $C$  is a flat  $A$ -algebra. Then use [Exercise 10.7](#) and [Exercise 3.17](#).

**10.9.** Assume that we have constructed  $g_k$  and  $h_k$  such that  $f - g_k h_k \in \mathfrak{m}^k[x]$ . Denote  $w_k = f - g_k h_k$ . Since there exist  $u_k$  and  $v_k$  such that  $1 - u_k g_k - v_k h_k \in \mathfrak{m}[x]$ , we have  $w_k - w_k u_k g_k - w_k v_k h_k \in \mathfrak{m}^{k+1}[x]$ .

Applying polynomial division algorithm, we have  $w_k u_k = q_k h_k + a_k$  with  $\deg a_k < \deg h_k$ . Since  $h_k$  monic, the division process shows that  $q_k \in \mathfrak{m}^k[x]$ , so  $a_k \in \mathfrak{m}^k[x]$ . Let  $b_k = w_k v_k + q_k g_k$ , then  $b_k \in \mathfrak{m}^k[x]$ . Note that  $\deg w_k < n$ , we have  $\deg b_k < \deg g_k$  in the polynomial ring  $(A/\mathfrak{m}^{k+1})[x]$ .

Let  $g_{k+1} = g_k + b_k$ ,  $h_{k+1} = h_k + a_k$ , then  $f - g_{k+1} h_{k+1} \in \mathfrak{m}^{k+1}[x]$ . Since  $A$  complete, the coefficients of  $g_k$  and  $h_k$  converge, so we have  $g, h \in A[x]$  such that  $g \equiv g_k \pmod{\mathfrak{m}^k}$  and  $h \equiv h_k \pmod{\mathfrak{m}^k}$ . Therefore,  $f - gh \in \bigcap_{k=1}^{\infty} \mathfrak{m}^k[x]$ . By [Proposition 10.17](#),  $\bigcap_{k=1}^{\infty} \mathfrak{m}^k[x] = 0$ .

**10.10.** (i)  $\bar{f}(x) = (x - \alpha)\bar{g}(x)$ .

(ii)  $\mathbb{Z}_7/7\mathbb{Z}_7 = \mathbb{Z}/7\mathbb{Z}$  by [Proposition 10.15\(iii\)](#). we also have  $(x^2 - \bar{2}) = (x + \bar{3})(x - \bar{3})$  in  $\mathbb{Z}/7\mathbb{Z}$ .

(iii)  $A = k[[x]]$  is a local ring with  $\mathfrak{m} = (x)$ .  $\bar{f}(\bar{0}, y) = (y - \bar{a})\bar{g}(\bar{0}, y) \in A[y]$ , so  $a \in A$ . Lift back by [Exercise 10.9](#) and let  $y(x) = a$ .

**10.11.** Let  $A$  be the ring of germs of  $C^\infty(\mathbb{R})$  functions. If  $f(0) \neq 0$ , then  $f(x)$  is always not 0 in  $A$ , so  $\mathfrak{m} = \{f(x) \in A : f(0) = 0\}$  is its only maximal ideal.

$\mathfrak{m}^n = \{f(x) \in A : f^{(0)}(0) = \dots = f^{(n-1)}(0) = 0\}$ , so  $A/\mathfrak{m}^n \cong \mathbb{R}[x]/(x^n)$  and therefore  $\hat{A} = \mathbb{R}[[x]]$ . By Borel's theorem,  $A \rightarrow \hat{A}$  is surjective, so  $\hat{A}$  is a finitely generated  $A$ -module.

If  $A$  is Noetherian, by [Proposition 10.17](#),  $\bigcap_{n=1}^{\infty} \mathfrak{m}^n = 0$ , but  $e^{-1/x^2} \in \bigcap_{n=1}^{\infty} \mathfrak{m}^n$ .

**10.12.** By [Proposition 10.14](#),  $A[[x_1, \dots, x_n]]$  is flat over  $A[x_1, \dots, x_n]$ , but from [Exercise 2.5](#) and [Exercise 2.8\(ii\)](#),  $A[x_1, \dots, x_n]$  is flat over  $A$ . Then use [Exercise 2.8\(ii\)](#) again.

## 11 Dimension Theory

**11.1.** Move  $P$  to the origin to remove the constant term of  $f$ .  $\mathfrak{m} = (x_1, \dots, x_n)/(f)$ , so  $\mathfrak{m}/\mathfrak{m}^2 = (x_1, \dots, x_n)/((x_1, \dots, x_n)^2 + (f))$ .  $\{x_i\}_{i=1}^n$  is a set of generators  $\mathfrak{m}/\mathfrak{m}^2$ . If  $f \in (x_1, \dots, x_n)^2$ ,  $\{x_i\}_{i=1}^n$  is a basis. If  $f \notin (x_1, \dots, x_n)^2$ , the existence of linear term of  $f$  means  $\{x_i\}_{i=1}^n$  is linearly dependent.

Use [Corollary 11.18](#) on  $A_{\mathfrak{m}} = k[x_1, \dots, x_n]_{\mathfrak{m}}/(f)_{\mathfrak{m}}$ , then [Theorem 11.22](#).

**11.2.** The map  $k[t_1, \dots, t_d]/(t_1, \dots, t_d)^n \rightarrow A/(x_1, \dots, x_d)^n$  is injective. Use [Corollary 7.16\(iii\)](#) and a similar step as [Exercise 10.5](#) to prove that  $\mathfrak{m}$ -adic topology and  $(x_1, \dots, x_d)$ -adic topology are same.

From [Proposition 10.22\(ii\)](#),  $G_{(t_1, \dots, t_d)}(k[[t_1, \dots, t_d]]) \cong G_{(t_1, \dots, t_d)}(k[t_1, \dots, t_d]) \cong k[t_1, \dots, t_d]$ . Note that every homogeneous element in  $G_{(x_1, \dots, x_d)}(A)$  has a preimage in  $k[t_1, \dots, t_d]$ , so  $G_{(x_1, \dots, x_d)}(A)$  is a finitely generated  $k[t_1, \dots, t_d]$ -module. Then use [Proposition 10.24](#).

**11.3.** Let  $\dim V = d$ . From [Noether normalization lemma](#),  $A(V)$  integral over  $B = k[x_1, \dots, x_d]$ . Since  $B$  is UFD, it is integrally closed. Apply [Exercise 5.3](#) on  $k \otimes_k k[x] \rightarrow \bar{k} \otimes_k k[x]$ , we also have  $C = \bar{k}[x_1, \dots, x_d]$  integral over  $B$ .

Let  $\mathfrak{m}$  be a maximal ideal in  $A(V)$ . By [Theorem 5.10](#),  $\mathfrak{m} \cap B$  is maximal in  $B$ , and can be lifted to a prime ideal  $\mathfrak{p}$  in  $C$ , and  $\mathfrak{p}$  is contained in a maximal ideal  $\mathfrak{n}$ , with  $\mathfrak{n} \cap B = \mathfrak{m} \cap B$ . If there is a chain in  $A(V)$  which ends at  $\mathfrak{m}$ , intersect with  $B$  and use [Theorem 5.16](#) to lift to  $C$ , we have a chain ends at  $\mathfrak{n}$ . Therefore,  $\dim C_{\mathfrak{n}} \geq \dim A(V)_{\mathfrak{m}}$ . Note that  $\dim C_{\mathfrak{n}} = d$  by [Theorem 11.25](#).

**11.4.** The maximal ideals in  $S^{-1}A$  is  $S^{-1}\mathfrak{p}_i$ . Localize at  $S^{-1}\mathfrak{p}_i$  we have a ring  $K[x_{m_i+1}, \dots, x_{m_{i+1}}]$ , where  $K = k[x_1, x_1^{-1}, \dots]$  without indeterminates  $x_{m_i+1}, \dots, x_{m_{i+1}}$ , so it is a field. Use [Exercise 7.9](#).

**11.5.** Let  $\gamma : (M) \rightarrow K(A_0)$ . Define the Poincaré series of  $M$  to be

$$P(M, t) = \sum_{n=0}^{\infty} \gamma(M_n) t^n \in K(A_0)[[t]].$$

**11.6.** From [Exercise 4.7\(ii\)](#), if  $\mathfrak{p}_0 \subsetneq \dots \subsetneq \mathfrak{p}_n$  is a chain in  $A$ , then  $\mathfrak{p}_0[x] \subsetneq \dots \subsetneq \mathfrak{p}_n[x] \subsetneq \mathfrak{p}_n[x] + (x)$  is a chain in  $A[x]$ .

Let  $f : A \rightarrow A[x]$  be the embedding. The fiber of  $f^*$  at  $\mathfrak{p}$  is  $\text{Spec}(k(\mathfrak{p}) \otimes_A A[x]) = \text{Spec}(k(\mathfrak{p})[x])$ . Since  $\dim(k(\mathfrak{p})[x]) = 1$ , a chain in  $A[x]$  whose contraction is  $\mathfrak{p}$  contains no more than 2 primes, so a chain in  $A[x]$  contains at most  $2 + 2 \dim A$  prime ideals.

**11.7.** Localize at  $\mathfrak{p}$ , and denote  $r = \text{height } \mathfrak{p} = \dim A_{\mathfrak{p}}$ . By [Theorem 11.14](#), there is a  $\mathfrak{p}$ -primary ideal  $\mathfrak{q}$  generated by  $r$  elements, so  $\mathfrak{q}[x]$  also generated by  $r$  elements. From [Exercise 4.7\(iii\)](#),  $\mathfrak{q}[x]$  is  $\mathfrak{p}[x]$ -primary, then use [Theorem 11.14](#) again we have  $\text{height } \mathfrak{p}[x] \leq r$ . Then use [Exercise 11.6](#).